COMPUTER SCIENCE TEXTBOOK SERIES

FOUNDATIONS OF
COMPUTER SCIENCE

■■■ FROM BASICS TO ARTIFICIAL INTELLIGENCE ■■■

✦ 19 Comprehensive Chapters ✦ Hardware · Software · OS · MS Office
Internet · Cyber Safety · Algorithms · Flowcharts · Scratch · Python
✦ Artificial Intelligence ✦ Ethics of AI

❮ 01010011 01000011 ❯

" Designed for First-Time Learners — No Prior Knowledge Required "

— BY —

The Monotheist

✦ ✦ ✦

Computer Science · Beginner Level · 19 Chapters

From Introduction to Computers → All the way through to → Artificial Intelligence & Ethics

TABLE OF CONTENTS

FRONT MATTER

Preface 3

UNIT 1 — COMPUTER FUNDAMENTALS

Ch. 1 Introduction to Computers 1

Ch. 2 Computer Hardware 12

Ch. 3 Computer Software 22

Ch. 4 Data Storage and Memory 32

Ch. 5 Operating System Basics 42

Ch. 6 File Management 52

UNIT 2 — PRODUCTIVITY & OFFICE TOOLS

Ch. 7 MS Word 62

Ch. 8 MS PowerPoint 74

Ch. 9 MS Excel 86

UNIT 3 — INTERNET, COMMUNICATION & SAFETY

Ch. 10 Internet Basics 98

Ch. 11 Email and Online Communication 108

Ch. 12 Cyber Safety and Digital Etiquette 118

UNIT 4 — COMPUTATIONAL THINKING & PROBLEM SOLVING

Ch. 13 Logical Thinking and Problem Solving 128

Ch. 14 Algorithms 138

Ch. 15 Flowcharts 148

UNIT 5 — INTRODUCTION TO CODING

Ch. 16 Introduction to Coding (Scratch) 158

Ch. 17 Basic Python Programming 168

UNIT 6 — ARTIFICIAL INTELLIGENCE & ETHICS

Ch. 18 Artificial Intelligence 180

Ch. 19 Ethics of AI 192

PREFACE

By The Monotheist

" In the name of God, the Most Gracious, the Most Merciful. "

Every journey of knowledge begins with a single step. This book is that step.

Why This Book Was Written

We live in an age where computers are no longer a luxury or a mystery reserved for scientists and engineers. They are present in our schools, hospitals, markets, and hands — and increasingly in the very decisions that shape our lives. Yet for millions of students, the door to understanding this world remains firmly closed, not because they lack ability, but because they have never been given the right key.

This book was written with one purpose: to be that key. It is designed for the student who has never touched a keyboard in a classroom, for the young mind full of curiosity but without guidance, and for the teacher who wants a single, complete, honest resource they can place in front of a beginner and say: "Start here."

I am not a publisher. I am not a corporation. I am a teacher — a person who believes that knowledge is a trust, that education is an act of worship, and that every child, regardless of where they come from, deserves to understand the world they are growing up in.

What You Will Find in This Book

This textbook covers nineteen carefully sequenced chapters, beginning with the most fundamental question — "What is a computer?" — and building steadily, step by step, all the way to Artificial Intelligence and the ethical questions it raises for humanity.

You will learn how computers are built and how they think. You will learn to work with the world's most widely used software tools: Microsoft Word, PowerPoint, and Excel. You will learn how the internet works, how to stay safe online, and how to communicate professionally in the digital world.

You will also learn to think logically, to design algorithms, to draw flowcharts, and to write your first real programs in Scratch and Python. In the final chapters, you will explore one of the most important conversations of our time: what Artificial Intelligence is, how it works, and — most importantly — what responsibilities come with building and using it.

How to Use This Book

Each chapter is designed to stand on its own while also building on what came before. Every chapter opens with Learning Objectives so you always know exactly what you are about to learn. The main content is explained in plain, clear language with real-world examples, illustrated tables, and highlighted concept boxes.

Every chapter closes with a full set of review questions — short answers, fill-in-the-blanks, true or false, and applied exercises. These are not tricks. They are tools. Use them honestly, and they will tell you exactly what you have learned and what still needs attention. A glossary of key terms is included in every chapter so that no word is left unexplained.

A Note on Artificial Intelligence

You will notice that this book ends with two chapters on Artificial Intelligence — one on how it works, and one on its ethics. This is intentional and important.

AI is not a distant future technology. It is already making decisions about what you see online, whether you get a bank loan, how doctors diagnose illness, and how conflicts may one day be resolved. Understanding AI — and being able to think critically about it — is not optional for the generation growing up today. It is a civic responsibility.

A Personal Word

I chose to write this book under the name The Monotheist as a reminder to myself — and to every reader — that all knowledge ultimately points to something greater than ourselves. The ability to reason, to create tools, to teach and to learn: these are gifts. And gifts, in the truest sense, carry obligations.

The obligation here is simple: use this knowledge well. Be honest. Be curious. Be kind to others in digital spaces as you would be in person. Build things that help people. Question things that harm them. And never stop learning.

To every student opening this book for the first time: I wrote it for you. I hope it opens doors.

With gratitude and sincerity,

The Monotheist

Author · Educator · Lifelong Student

CHAPTER 1

INTRODUCTION TO COMPUTERS

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Define a computer and explain what it does
- ✓ Identify and describe the main parts of a computer
- ✓ Understand hardware and software with examples
- ✓ Explain how data is stored and measured
- ✓ Describe different types and generations of computers
- ✓ Understand the basic concept of networking and the internet
- ✓ Identify real-world uses of computers

1.1 What is a Computer?

A computer is an electronic device that accepts data as input, processes it according to a set of instructions (called a program), and produces useful information as output. It can store data for future use and perform tasks very quickly and accurately.

The word "computer" originally referred to a person who performed calculations. Over time, machines were built to do this work, and today a computer can perform millions of tasks in just one second.

Key Characteristics of a Computer

Characteristic	Description
Speed	Computers perform millions of operations per second.
Accuracy	They produce correct results when given correct instructions.
Storage	They can store vast amounts of data permanently or temporarily.
Automation	Once programmed, they work without human intervention.
Diligence	They never get tired and work consistently around the clock.
Versatility	They can perform many different kinds of tasks.

■ Did You Know?

The first general-purpose electronic computer, ENIAC, was built in 1945. It weighed about 30 tons and filled an entire room. Today, your smartphone is millions of times more powerful!

1.2 The IPO Cycle (Input–Process–Output)

Every computer works on the IPO (Input–Process–Output) cycle. This is the most fundamental concept in computing:



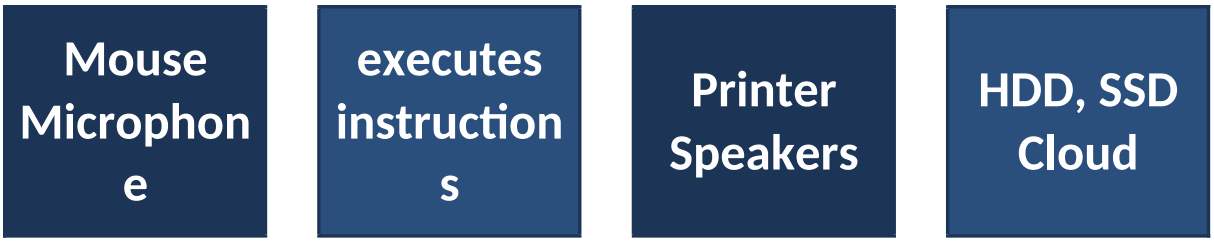


Figure 1.1: The IPO Cycle — how every computer processes information

Stage	What Happens	Example
INPUT	Data is entered into the computer	Typing on a keyboard, clicking a mouse
PROCESS	The CPU works on the data using instructions	CPU calculates, compares, or transforms the data
OUTPUT	Results are shown to the user	Displaying text on screen, printing a document
STORAGE	Data is saved for later use	Saving a file to the hard disk

1.3 Hardware

Hardware refers to all the physical parts of a computer — the parts you can see and touch. Hardware is divided into four main categories: input devices, output devices, the processing unit (CPU), and memory.

1.3.1 Input Devices

Device	Description
Keyboard	Used to type letters, numbers, and symbols. The most common input device.
Mouse	A pointing device used to click, drag, and select items on screen.
Scanner	Converts printed documents or images into digital form.
Microphone	Captures sound and converts it to digital audio.
Webcam	Captures live video and sends it to the computer.
Touchscreen	Allows input by touching the screen directly.
Joystick	Used mainly for games to control movement.
Barcode Reader	Scans barcodes to read product information.

1.3.2 Output Devices

Device	Description
Monitor	The screen that displays text, images, and video.
Printer	Produces a paper (hard) copy of documents or images.
Speakers	Produce sound — music, speech, or alerts.
Projector	Projects the screen image onto a large surface or wall.
Headphones	Personal audio output worn on or over the ears.
Plotter	Draws large-scale diagrams or blueprints (used in engineering).

1.3.3 Processing Unit — The CPU

The Central Processing Unit (CPU) is often called the "brain" of the computer. It carries out all instructions given to the computer.

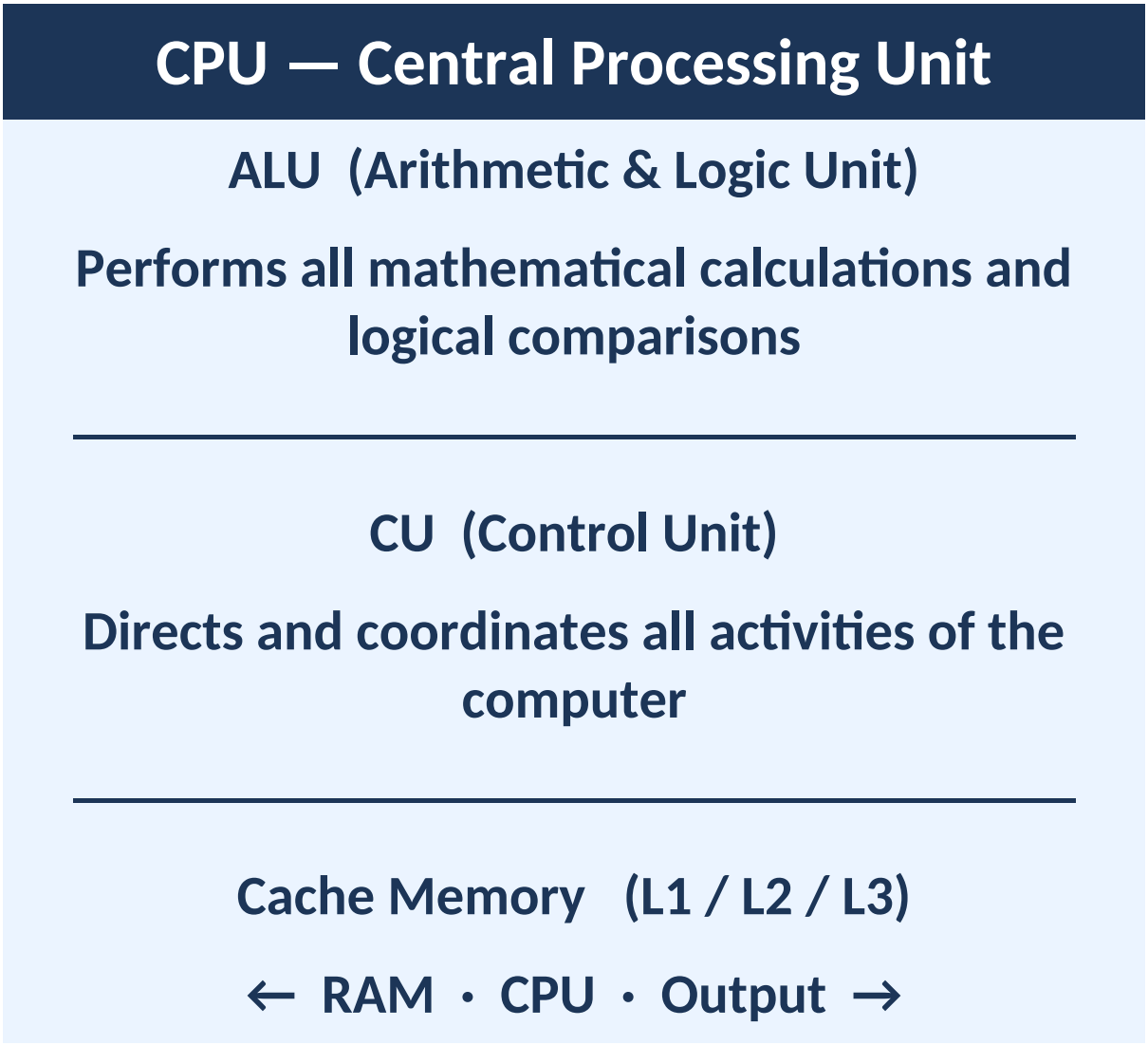


Figure 1.2: Internal structure of the CPU

CPU speed is measured in GHz (Gigahertz). A 3.0 GHz processor can perform 3 billion cycles per second. Modern computers may also have multiple cores (dual-core, quad-core, etc.), allowing them to work on several tasks simultaneously.

1.3.4 Memory

■ RAM vs. ROM

RAM (Random Access Memory): Temporary storage used while the computer is running. It is fast but loses all data when power is off.

ROM (Read-Only Memory): Permanent memory containing the computer's basic startup instructions (BIOS/UEFI). It retains data even when power is off.

Storage Device	Description
Hard Disk Drive (HDD)	Magnetic storage. Common in desktops and laptops. Capacity: 500 GB to several TB.
Solid State Drive (SSD)	Faster and more durable than HDD. Uses flash memory chips. No moving parts.

Storage Device	Description
USB Flash Drive	Small, portable storage device. Easy to carry and use.
CD / DVD	Optical discs. CDs hold ~700 MB; DVDs hold 4.7 GB to 17 GB.
Memory Card	Small cards used in cameras, phones, and tablets (SD card, microSD).
Cloud Storage	Storing files online on remote servers (e.g., Google Drive, OneDrive).

1.4 Data Storage Units

Unit	Size / Description
Bit (b)	The smallest unit. Value is either 0 or 1.
Nibble	4 bits
Byte (B)	8 bits. Can store one character (e.g., the letter "A").
Kilobyte (KB)	1,024 Bytes — approximately one page of text.
Megabyte (MB)	1,024 KB — approximately one small photo.
Gigabyte (GB)	1,024 MB — approximately 1,000 large photos.
Terabyte (TB)	1,024 GB — approximately 500 hours of HD video.
Petabyte (PB)	1,024 TB — used in large data centres.

Did You Know?

A typical song is about 4 MB. A full HD movie is about 4 GB. A computer with 1 TB of storage can hold approximately 250,000 songs!

1.5 Software

Software is the set of instructions (programs) that tells the hardware what to do. Unlike hardware, you cannot touch software. Software is divided into two major categories: system software and application software.

Category	Examples
Word Processing	Microsoft Word, Google Docs, LibreOffice Writer
Spreadsheet	Microsoft Excel, Google Sheets
Presentation	Microsoft PowerPoint, Google Slides
Web Browser	Google Chrome, Mozilla Firefox, Microsoft Edge
Multimedia	VLC Media Player, Adobe Photoshop, Audacity
Communication	WhatsApp, Zoom, Microsoft Teams, Gmail
Education	Khan Academy, Duolingo, Google Classroom
Games	Chess, Minecraft, FIFA

1.6 Types of Computers





Figure 1.3: Types of computers — from most powerful (left) to most portable (right)

Type	Description
Supercomputer	The fastest and most powerful computers. Used for weather forecasting, nuclear research, space exploration. Example: Frontier (USA).
Mainframe	Very large and powerful. Used by banks, airlines, governments to process millions of transactions daily.
Personal Computer (PC)	Designed for individual use at home or office. Includes desktops and laptops.
Laptop / Notebook	A portable PC with built-in keyboard, screen, and battery.
Tablet	A flat touchscreen device. Examples: iPad, Samsung Galaxy Tab.
Smartphone	A pocket-sized computer with phone capabilities.
Embedded Computer	Built inside other devices. Examples: washing machines, microwaves, ATMs.
Wearable Computer	Worn on the body. Examples: smartwatch, fitness tracker.

1.7 Generations of Computers

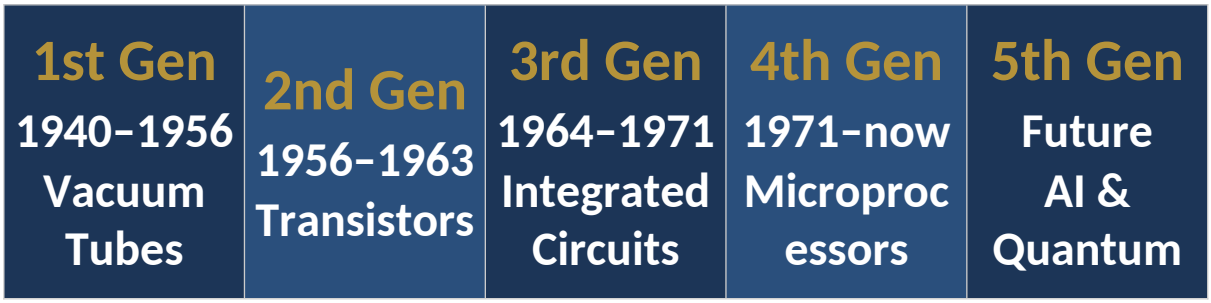


Figure 1.4: The five generations of computers

Generation	Period & Technology	Key Features
1st	1940–1956 | Vacuum Tubes	Very large, consumed huge power, produced a lot of heat. Machine language. ENIAC, UNIVAC.
2nd	1956–1963 | Transistors	Smaller, faster, more reliable. Assembly language. IBM 1401.
3rd	1964–1971 | Integrated Circuits	Even smaller and faster. High-level languages. IBM System/360.
4th	1971–present | Microprocessors	Entire CPU on a single chip. Birth of personal computers. Intel processors.
5th	Future | AI & Quantum	Machines that can think and learn. AI assistants, quantum computers.

1.8 Computer Networks and the Internet



Figure 1.5: Types of computer networks by geographic coverage

Network Type	Coverage	Example Use
LAN (Local Area Network)	Small area: classroom, office, home	School computer lab network
WAN (Wide Area Network)	Large area: cities, countries, world	The Internet (largest WAN)
MAN (Metropolitan Area Network)	City or large campus	City-wide Wi-Fi, university network
WLAN (Wireless LAN)	LAN using Wi-Fi instead of cables	Home Wi-Fi network

■ Web vs. Internet

The Internet is the global physical network of cables and connections.

The World Wide Web (WWW) is a service that runs ON the Internet. Not all internet services are the Web — email and online gaming also use the internet but are not part of the WWW.

1.9 Uses of Computers

Field	How Computers Are Used
Education	E-learning, digital classrooms, research, online libraries
Healthcare	Patient records, diagnosing diseases, medical imaging (X-ray, MRI)
Banking & Finance	Online banking, ATMs, stock trading, financial records
Communication	Email, video calls, social media, instant messaging
Entertainment	Games, streaming movies and music, digital art
Science & Research	Simulations, data analysis, weather forecasting
Government	Record keeping, e-governance, law enforcement
Business	Inventory management, payroll, marketing, e-commerce
Agriculture	Crop monitoring, weather prediction, automated farming
Transportation	GPS navigation, traffic management, booking systems

1.10 Advantages and Disadvantages of Computers

Advantages	Disadvantages
Very fast at processing large amounts of data.	Requires electricity — useless during power cuts without backup.
High accuracy — no mistakes when given correct input.	Can malfunction due to viruses, software bugs, or hardware failure.
Can store enormous amounts of data in a small space.	Over-reliance can reduce critical thinking skills.
Capable of multitasking — running many programs at once.	Privacy and security risks — hacking, data theft.
Available 24/7 without fatigue.	Health issues from prolonged use — eye strain, poor posture.
Enables global communication and access to information.	Expensive to purchase, maintain, and upgrade.

Chapter Review Questions

A. Short Answer Questions

- What is a computer? List any three characteristics.
- Explain the IPO cycle with an example.
- What is the difference between RAM and ROM?
- Name four input devices and four output devices.
- What is the difference between hardware and software?
- Name the five generations of computers and the technology used in each.
- What is a LAN? How is it different from a WAN?
- List any five uses of computers in daily life.

B. Fill in the Blanks

- The _____ is called the brain of the computer.
- 8 bits make one _____.
- _____ is temporary memory that is lost when the computer is turned off.
- The _____ is the largest WAN in the world.
- The first generation of computers used _____ as their main technology.

C. True or False

- A printer is an input device. (True / False)
- ROM retains data even when the computer is off. (True / False)
- A Supercomputer is the slowest type of computer. (True / False)
- The Internet and the World Wide Web are exactly the same thing. (True / False)
- Software can be physically touched. (True / False)

Key Terms Glossary

Term	Definition
Computer	An electronic device that inputs, processes, stores, and outputs data.
Hardware	Physical components of a computer that can be seen and touched.
Software	Programs and instructions that tell hardware what to do.

CPU	Central Processing Unit — the brain of the computer.
RAM	Random Access Memory — fast, temporary memory used while the computer is on.
ROM	Read-Only Memory — permanent memory storing startup instructions.
Input Device	Device used to enter data into the computer (e.g., keyboard, mouse).
Output Device	Device that delivers processed results to the user (e.g., monitor, printer).
Network	A group of computers connected to share resources and information.
Internet	A global network connecting billions of computers worldwide.
Bit	The smallest unit of data; a single 0 or 1.
Byte	8 bits; the basic unit used to measure data storage.
Gigabyte (GB)	1,024 Megabytes; common unit for measuring storage.
Application Software	Programs designed for specific user tasks (e.g., Word, Chrome).
Embedded Computer	A computer built inside another device (e.g., in a microwave or car).

— End of Chapter 1 —

CHAPTER 2

COMPUTER HARDWARE

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Identify and describe all major internal and external hardware components
- ✓ Explain the role of the motherboard in connecting computer components
- ✓ Understand the CPU, GPU, RAM, and their functions in detail
- ✓ Describe common ports and connectors and their uses
- ✓ Distinguish between internal and external hardware
- ✓ Explain how hardware components work together as a system

2.1 What is Computer Hardware?

Computer hardware refers to all the physical, tangible parts of a computer system — the components you can see, touch, and handle. Hardware can be divided into two main categories: internal hardware (components inside the computer case) and external hardware (devices outside the case, also called peripherals).

2.2 The Motherboard

The motherboard is the main circuit board of a computer. It is the central hub that connects all other hardware components together, allowing them to communicate with each other.

Component	Description
CPU Socket	The slot where the processor (CPU) is installed.
RAM Slots	Slots where RAM (memory) modules are inserted.
BIOS/UEFI Chip	Small chip containing the firmware that starts the computer.
Expansion Slots (PCIe)	Slots for adding graphics cards, sound cards, and other expansion cards.
Storage Connectors (SATA/M.2)	Ports for connecting hard drives (HDD) and solid-state drives (SSD).
Power Connector	Receives power from the power supply unit (PSU).
Chipset	Controls data flow between the CPU, RAM, and other components.
I/O Ports	USB, HDMI, audio, and other ports on the back panel for peripherals.

■ Did You Know?

Modern motherboards use UEFI (Unified Extensible Firmware Interface) instead of the older BIOS. UEFI supports faster boot times, larger hard drives, and a graphical interface.

2.3 The Central Processing Unit (CPU)

Factor	Explanation
Clock Speed (GHz)	How many instruction cycles the CPU completes per second. Higher = faster.
Number of Cores	Each core can process tasks independently. Quad-core CPUs handle 4 tasks at once.

Factor	Explanation
Cache Memory	Ultra-fast memory built into the CPU (L1, L2, L3). Stores frequently used data.
Architecture (nm)	Smaller nanometer size means more efficient processing and less heat.
Threads	Virtual cores. A CPU with 4 cores and 8 threads handles 8 simultaneous tasks.

2.4 Graphics Processing Unit (GPU)

Type	Description	Example Uses
Integrated GPU	Built into the CPU. Suitable for everyday tasks.	Office work, web browsing, video playback
Dedicated GPU	Separate card on the motherboard. Much more powerful.	Gaming, video editing, 3D design, AI training

2.5 RAM (Random Access Memory)

RAM Amount	Suitable For
4 GB	Basic tasks: browsing, documents, light multitasking.
8 GB	Everyday use: multitasking, light gaming, most student and office needs.
16 GB	Demanding tasks: video editing, gaming, programming.
32 GB or more	Professional work: 3D rendering, large databases, AI development.

■ RAM vs. Storage

RAM is NOT the same as storage. RAM holds data temporarily while programs are running — it is erased when the computer turns off.

Storage (HDD or SSD) holds data permanently — your files, photos, and programs are stored there even when the computer is off.

2.6 Power Supply Unit (PSU)

The Power Supply Unit (PSU) converts AC power from the wall socket into the DC power that computer components need. PSU power is measured in watts (W). A basic office computer may need 300W, while a high-performance gaming computer may require 750W or more.

2.7 Cooling Systems

Cooling Type	How It Works	Best For
Air Cooling (Fans)	Fans push heat away from components using heatsinks.	Most desktops and laptops
Liquid Cooling	Water or coolant absorbs heat and carries it to a radiator.	High-performance gaming and workstations
Thermal Paste	Conducts heat from the CPU to the heatsink efficiently.	Applied between CPU and cooler
Laptop Cooling Pad	External fan placed under a laptop to increase airflow.	Laptops during intensive use

2.8 Ports and Connectors

Port	Use
USB-A	Connects keyboards, mice, flash drives, and many other devices.
USB-C / Thunderbolt	High-speed data transfer, charging, and video output. Modern standard.
HDMI	Connects monitors, TVs, and projectors for video and audio output.
DisplayPort	High-resolution video output, common in gaming monitors.
3.5mm Audio Jack	Connects headphones, speakers, and microphones.
Ethernet (RJ-45)	Wired internet connection. Faster and more reliable than Wi-Fi.
SD Card Slot	Reads memory cards from cameras and other devices.
VGA	Older video output port, still found on some projectors and monitors.

2.9 How Hardware Works Together

- The CPU reads the program's instructions from the storage drive (HDD/SSD).
- The program is loaded into RAM so the CPU can access it quickly.
- As you type, the keyboard (input device) sends signals to the CPU.
- The CPU processes your keystrokes and updates the document in RAM.
- The GPU renders the updated text and sends it to the monitor (output device).
- When you save, the CPU writes data from RAM back to storage.

Chapter Review Questions

A. Short Answer Questions

- What is a motherboard? List four components found on it.
- What does GHz measure in a CPU? What does having multiple cores mean?
- What is the difference between a CPU and a GPU?
- Why is RAM important? How does more RAM help a computer?
- What is a PSU? Why is it essential to a computer system?
- Name four types of ports and explain what each connects.
- What is the difference between internal hardware and external hardware (peripherals)?

B. Fill in the Blanks

- The _____ is the main circuit board that connects all computer components.
- CPU speed is measured in _____ (GHz).
- The _____ converts AC power from the wall into DC power for the computer.
- A dedicated _____ (GPU) is used for gaming and video editing.
- Ports like USB and HDMI are found on the _____ panel of the computer.

C. True or False

- RAM permanently stores your files even when the computer is off. (True / False)
- A quad-core CPU can process four tasks simultaneously. (True / False)
- Liquid cooling is more effective than air cooling for high-performance computers. (True / False)
- HDMI is used to connect audio headphones to a computer. (True / False)
- The motherboard connects all hardware components together. (True / False)

Key Terms Glossary

Term	Definition
Motherboard	The main circuit board that connects and coordinates all computer components.
CPU	Central Processing Unit — executes program instructions and processes data.
GPU	Graphics Processing Unit — handles visual rendering and parallel computing tasks.
RAM	Temporary working memory used by the CPU while programs are running.
PSU	Power Supply Unit — converts mains electricity into power for computer components.
PCIe Slot	Expansion slot on the motherboard for adding graphics cards and other cards.
UEFI	Unified Extensible Firmware Interface — modern firmware replacing BIOS.
USB	Universal Serial Bus — standard port for connecting peripherals.
HDMI	High-Definition Multimedia Interface — port for video and audio output.
Peripheral	An external hardware device connected to the computer.

— End of Chapter 2 —

CHAPTER 3

COMPUTER SOFTWARE

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

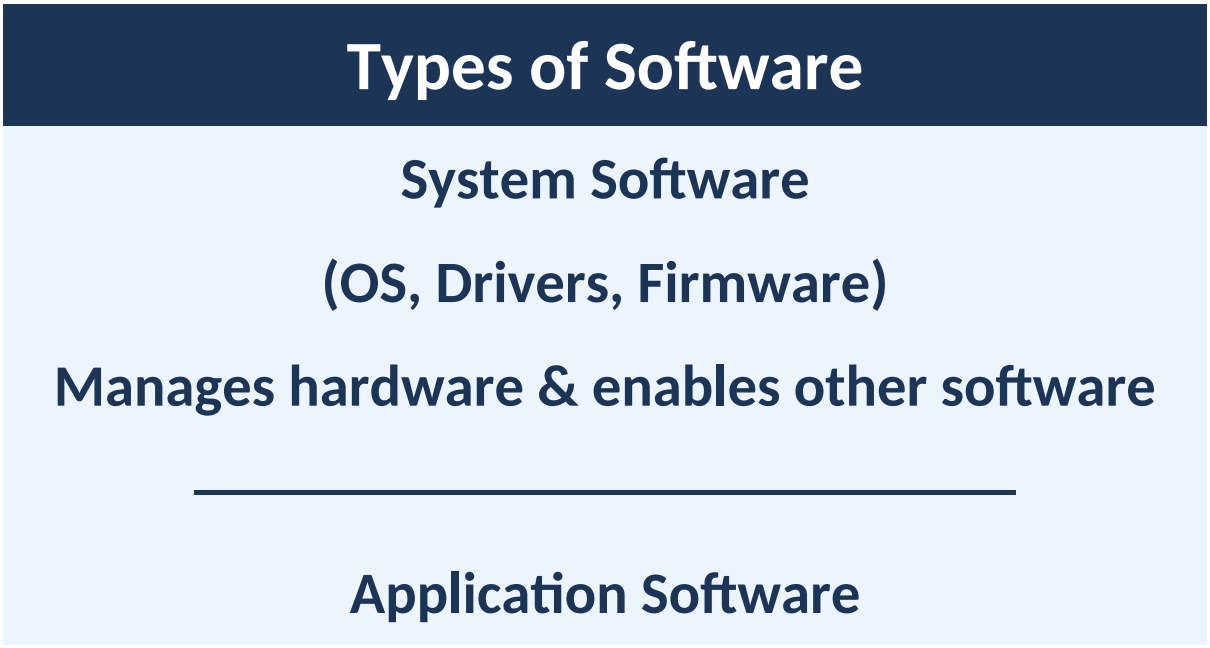
- ✓ Define software and explain the difference between software and hardware
- ✓ Distinguish between system software, application software, and utility software
- ✓ Identify examples of each type of software
- ✓ Understand how programming languages work
- ✓ Compare open-source and proprietary software
- ✓ Understand software licensing and how to install and update software

3.1 What is Software?

Software is the collection of programs, instructions, and data that tells a computer how to perform tasks. While hardware is the physical machinery, software is the "thinking" — the set of commands that makes hardware useful.

Feature	Hardware	Software
Physical?	Yes — you can touch it	No — it exists as code and data
Examples	CPU, keyboard, monitor	Windows, Word, Chrome
Can be damaged physically?	Yes	No (but can be corrupted)
Needs the other to work?	Yes — needs software to function	Yes — needs hardware to run

3.2 Types of Software



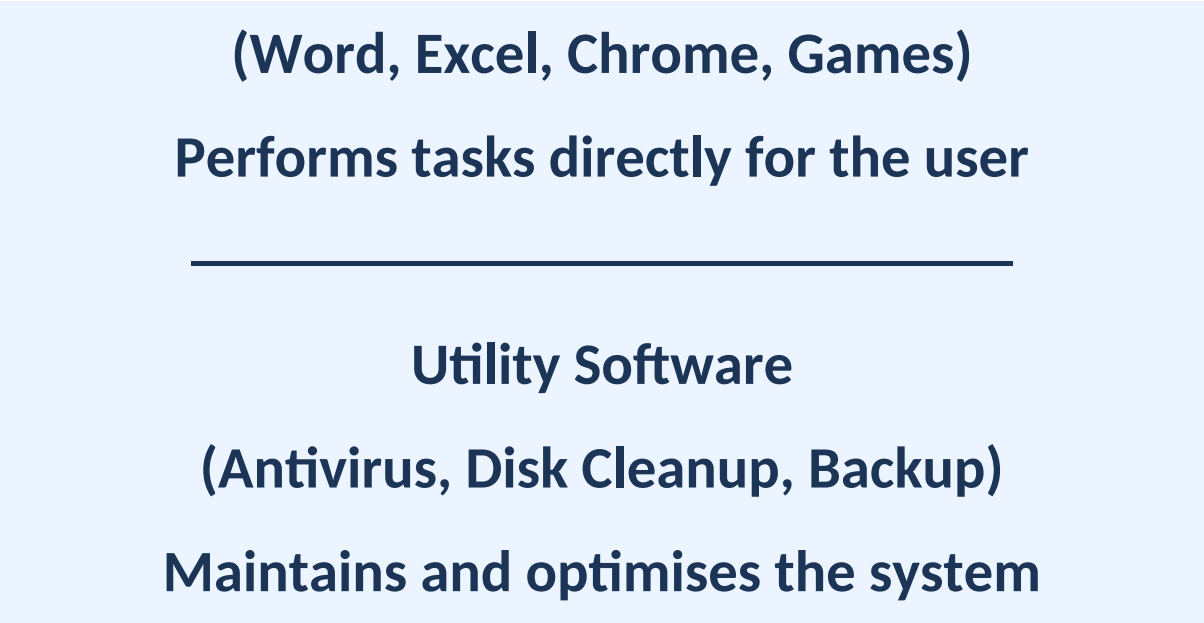


Figure 3.1: The three categories of software

3.3 System Software

System software manages the hardware and provides a platform for other software to run. It operates in the background and is essential for the computer to function.

3.3.1 Operating System (OS)

The Operating System is the most important piece of system software. It manages all hardware resources and allows application software to run. Examples include Windows, macOS, Linux, Android, and iOS.

3.3.2 Device Drivers

Device drivers are small programs that allow the operating system to communicate with specific hardware devices. Without a driver, the OS cannot use the device.

3.3.3 Firmware

Firmware is low-level software permanently stored in hardware chips. Examples include the BIOS/UEFI in a computer and the software embedded in a router.

3.4 Application Software

Category	Examples	Purpose
Word Processing	Microsoft Word, Google Docs	Creating and editing text documents
Spreadsheet	Microsoft Excel, Google Sheets	Organising and analysing numerical data
Presentation	Microsoft PowerPoint, Google Slides	Creating visual slideshows
Web Browser	Google Chrome, Mozilla Firefox, Safari	Accessing and browsing websites
Email Client	Gmail, Microsoft Outlook	Sending, receiving, and managing emails
Media Player	VLC, Windows Media Player, Spotify	Playing audio and video files
Image Editing	Adobe Photoshop, GIMP, Canva	Editing and creating images and graphics
Communication	WhatsApp, Zoom, Microsoft Teams	Video calls, messaging, and conferencing
Education	Khan Academy, Duolingo	Learning and teaching

3.5 Utility Software

Utility	Purpose
Antivirus / Anti-malware	Scans for and removes viruses, spyware, and malicious programs.
Disk Defragmenter	Rearranges fragmented files on an HDD for faster read/write access.
Disk Cleanup	Removes temporary files, cache, and unneeded data to free storage space.
File Compression (ZIP/RAR)	Reduces file sizes for easier storage and faster sharing.
Backup Software	Creates copies of important files to protect against data loss.
Task Manager	Shows CPU, RAM, and storage usage; allows closing frozen programs.
Firewall	Monitors and controls incoming and outgoing network traffic for security.

3.6 Programming Languages

Programming languages allow humans to write instructions that a computer can understand. High-level languages use English-like syntax and must be translated into machine code by a compiler or interpreter before the CPU can run them.

Language	Primary Use
Python	Beginner-friendly; used in AI, data science, web development, automation
JavaScript	Web development (interactive websites)
Java	Enterprise applications, Android development
C / C++	System programming, game development, high-performance software
HTML / CSS	Web page structure and styling
SQL	Querying and managing databases
Scratch	Visual block-based language for beginners and children

3.7 Open-Source vs. Proprietary Software

Feature	Open-Source Software	Proprietary Software
Source Code	Publicly available — anyone can view, modify, and share it.	Private — only the company has access.
Cost	Usually free.	Often requires purchase or subscription.
Examples	Linux, Firefox, VLC, GIMP, LibreOffice	Windows, Microsoft Office, Adobe Photoshop
Customisation	Highly customisable.	Limited customisation allowed.

3.8 Software Licensing

A software licence is a legal agreement specifying how software may be used. When you install software, you agree to its licence terms. Using software without the correct licence is illegal.

Licence Type	Description	Examples
Commercial / Retail	Buy once and use permanently on a set number of devices.	Older Microsoft Office
Subscription	Pay monthly or yearly to keep using the	Microsoft 365, Adobe Creative

Licence Type	Description	Examples
	software.	Cloud
Freeware	Free to use; source code not publicly available.	VLC, 7-Zip, Notepad++
Shareware	Free trial with limited features; pay for full version.	WinZip, some antivirus programs
Open-Source (GPL)	Free to use, modify, and distribute under the licence terms.	Linux, LibreOffice, Firefox
Proprietary	Paid and restricted — cannot be modified or redistributed.	Windows, macOS, Photoshop

3.9 Installing and Updating Software

Installing Software Safely

- Download from the official developer website or a trusted app store only.
- Run the installer and follow the on-screen instructions.
- Accept the licence agreement (read it — it is a legal contract).
- Choose the installation folder or use the default location.
- Restart the computer if prompted to complete installation.

Why Updates Matter

- Security patches — fix vulnerabilities that hackers could exploit.
- Bug fixes — correct errors or crashes in the software.
- New features — additions and improvements to the software.

■ Best Practice

Always download software from the official developer's website. Unofficial or pirated software can contain malware that damages your computer or steals your data. Enable automatic updates wherever possible.

Chapter Review Questions

A. Short Answer Questions

- What is software? How is it different from hardware?
- What is the difference between system software and application software?
- Name three types of utility software and explain what each does.
- What is the difference between machine language and a high-level language?
- What is a software licence? Name three different types of licence.
- What is open-source software? Give two advantages over proprietary software.
- Why is it important to keep software regularly updated?

B. Fill in the Blanks

- _____ software manages hardware and provides a platform for other software.
- Microsoft Windows and macOS are examples of _____ software.
- Open-source software makes its _____ code publicly available.
- A _____ licence requires you to pay monthly or yearly to use the software.
- A _____ translates high-level source code into machine language.

C. True or False

- Application software is the most important type of system software. (True / False)
- Python is a high-level programming language. (True / False)
- A device driver allows the OS to communicate with a specific hardware device. (True / False)

- Updating software regularly is important for security. (True / False)
- Using software without a valid licence is illegal. (True / False)

Key Terms Glossary

Term	Definition
Software	Programs and instructions that tell hardware how to perform tasks.
System Software	Software that manages hardware and supports other programs (e.g., OS, drivers).
Application Software	Software designed for specific user tasks (e.g., Word, Chrome, games).
Utility Software	Software that maintains and optimises a computer system.
Operating System	System software managing all hardware resources and running applications.
Device Driver	A program allowing the OS to communicate with a specific hardware device.
Compiler	A program that translates high-level source code into machine language for the CPU to execute.
Firmware	Low-level software stored permanently in hardware chips (e.g., BIOS/UEFI).
Open-Source	Software whose source code is publicly available to use, modify, and distribute.
Proprietary	Software whose source code is private and cannot be freely modified or shared.
Freeware	Software that is free to use but whose source code is not publicly available.

— End of Chapter 3 —

CHAPTER 4

DATA STORAGE AND MEMORY

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Explain the difference between primary and secondary storage
- ✓ Compare different storage devices and their characteristics
- ✓ Understand data storage units from bit to petabyte
- ✓ Describe how data is represented in binary
- ✓ Evaluate appropriate storage media for different situations

4.1 Why Do Computers Need Storage?

A computer needs to store two things: the programs it runs and the data those programs create. Without storage, a computer could not remember anything — not your files, your photos, or even how to start up. Storage keeps data safe for as long as needed, from a few seconds to many years.

Memory Hierarchy — Speed vs. Capacity

Registers (CPU) — Fastest / Smallest

↓

Cache L1 / L2 / L3 — Very Fast / Small

↓

RAM — Fast / Medium Capacity

↓

SSD / HDD — Slower / Large Capacity

↓

Cloud / Tape — Slowest / Huge Capacity

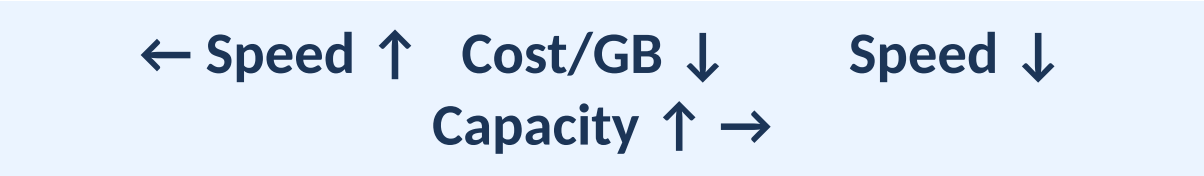


Figure 4.1: The memory hierarchy — faster storage at top, higher capacity at bottom

4.2 Primary Memory

Primary memory is directly accessible by the CPU and is used for active, real-time operations. It is fast but limited in capacity.

4.2.1 RAM (Random Access Memory)

RAM is the working memory of a computer. All programs and data currently in use are loaded into RAM. When you close a program or turn off the computer, the RAM contents are erased.

Property	Detail
Type	Volatile — loses data without power
Speed	Very fast (measured in MHz/GHz and bandwidth GB/s)
Capacity	Typically 4 GB – 64 GB in modern computers
Purpose	Holds the OS, open applications, and active data

4.2.2 ROM (Read-Only Memory)

ROM is a small chip on the motherboard containing the firmware — the initial instructions the computer needs to start up (BIOS or UEFI). It is permanent and cannot be changed by normal user operations.

4.2.3 Cache Memory

Cache memory is ultra-fast memory built directly into the CPU. It stores the most frequently used instructions and data so the CPU does not have to fetch them from the slower RAM. There are three levels:

Cache Level	Description
L1 Cache	Smallest and fastest. Built directly into the CPU core itself.
L2 Cache	Slightly larger and slower than L1. Per core or shared between cores.
L3 Cache	Largest cache level. Shared between all CPU cores.

4.3 Secondary Storage

Secondary storage is used for long-term, permanent storage. Data remains stored even when the computer is turned off. It is slower than RAM but far larger in capacity.

Type	Examples	Key Feature
Magnetic	HDD, Magnetic Tape	Affordable, high capacity, slower
Optical	CD, DVD, Blu-ray	Read-only or writable, durable
Flash / Solid-State	SSD, USB Drive, SD Card	Fast, silent, no moving parts
Cloud	Google Drive, OneDrive, Dropbox	Accessible anywhere, requires internet

■ HDD vs. SSD

HDD (Hard Disk Drive): Uses spinning magnetic platters. Affordable and high-capacity but slower and more fragile. Capacity: 500 GB – 10+ TB.

SSD (Solid State Drive): Uses flash memory chips with no moving parts. 5–10× faster than HDD, more durable, silent, but more expensive per GB.

4.4 Data Representation: Binary

All data in a computer is ultimately represented as binary — sequences of 0s and 1s. This is because electronic circuits have two states: off (0) and on (1).

Each bit position represents a power of 2 (from right: 1, 2, 4, 8, 16...). Example: 1011 = (1×8) + (0×4) + (1×2) + (1×1) = 8 + 0 + 2 + 1 = 11 in decimal.

Binary	Decimal Value
0000	0
0001	1
0010	2
0011	3
0100	4
0101	5
1000	8
1111	15

Character Encoding

Standard	Description
ASCII	American Standard Code for Information Interchange. 7-bit code for 128 characters. 'A' = 65, 'a' = 97, '0' = 48.
Unicode (UTF-8)	Modern standard supporting over 143,000 characters — all world languages, emojis, and mathematical symbols.

4.5 Storage Units — Full Breakdown

Unit	Size	Real-World Example
Bit (b)	Smallest unit — a single 0 or 1	A single switch in a circuit
Nibble	4 bits	Half a character
Byte (B)	8 bits — one character	One letter or digit
Kilobyte (KB)	1,024 bytes	Approximately one paragraph of text
Megabyte (MB)	1,024 KB	Approximately one digital photo
Gigabyte (GB)	1,024 MB	Approximately 250 songs
Terabyte (TB)	1,024 GB	Approximately 500 hours of HD video
Petabyte (PB)	1,024 TB	Used in large data centres
Exabyte (EB)	1,024 PB	Entire internet traffic per day

Chapter Review Questions

A. Short Answer Questions

- What is the difference between primary and secondary storage?
- Why is cache memory important? Name and describe the three levels of cache.
- Compare HDDs and SSDs in terms of speed, durability, and cost.
- What is binary? Why do computers use binary to represent data?
- What is the difference between ASCII and Unicode? Give an example of each.
- Name and describe three types of secondary storage devices.

B. Fill in the Blanks

- _____ memory is lost when the computer loses power.
- A group of 8 bits is called a _____.
- The binary number 1010 equals _____ in decimal.
- _____ storage saves data on remote servers accessed over the internet.
- ROM stores the _____ that starts up the computer.

C. True or False

- RAM is non-volatile memory. (True / False)
- A USB flash drive uses flash memory with no moving parts. (True / False)
- Blu-ray discs can store more data than DVDs. (True / False)
- Unicode only supports the English alphabet. (True / False)
- L1 cache is the largest and slowest cache level. (True / False)

Key Terms Glossary

Term	Definition
Volatile Memory	Memory that loses its data when power is removed (e.g., RAM).
Non-volatile Memory	Memory that retains data without power (e.g., ROM, SSD, HDD).
Cache Memory	Ultra-fast memory built into the CPU for frequently used data.
Binary	A number system using only 0 and 1 — the language of computers.
ASCII	A character encoding standard representing 128 characters as numbers.
Unicode	A modern encoding standard for all world languages and symbols.
Cloud Storage	Storing data on remote internet servers accessible from anywhere.
Flash Memory	Non-volatile storage using electronic cells — used in SSDs and USB drives.
HDD	Hard Disk Drive — magnetic storage with spinning platters; large capacity, slower.
SSD	Solid State Drive — flash-based storage; faster, silent, more durable than HDD.

— End of Chapter 4 —

CHAPTER 5

OPERATING SYSTEM BASICS

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Define an operating system and explain its purpose
- ✓ Describe the main functions of an operating system
- ✓ Compare major operating systems: Windows, macOS, Linux, Android, iOS
- ✓ Understand how the OS manages processes, memory, and user accounts
- ✓ Explain the role of the GUI and CLI
- ✓ Identify common utility programs and their purposes

5.1 What is an Operating System?

An operating system (OS) is system software that manages all the hardware resources of a computer and provides a platform for other software to run. Without an OS, a computer cannot function. The OS acts as an intermediary between the user, the applications, and the hardware.

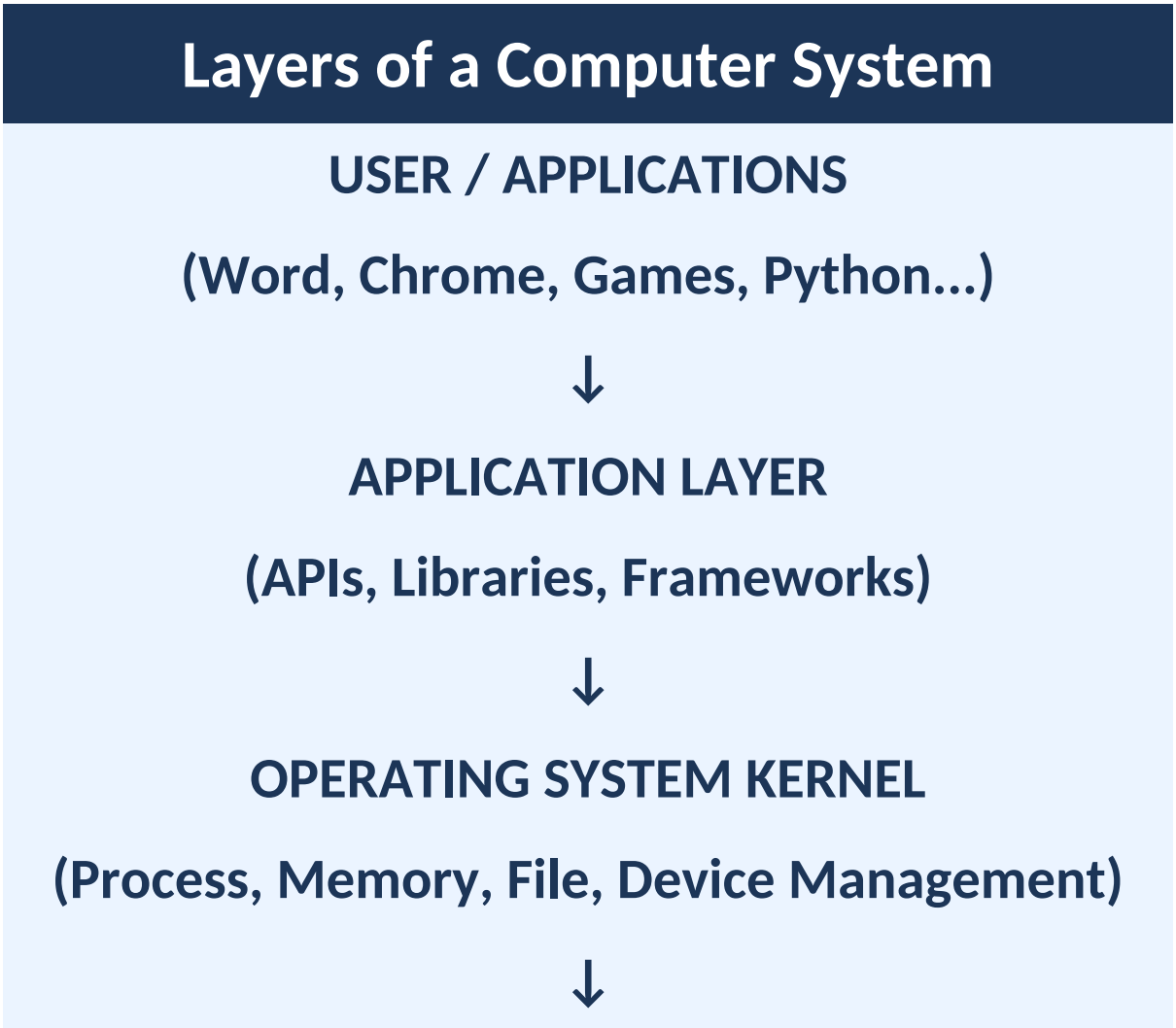




Figure 5.1: The OS sits between hardware and applications

5.2 Key Functions of an Operating System

Function	Description
Process Management	Decides which programs run, shares CPU time, and handles multitasking.
Memory Management	Allocates RAM to running programs and frees it when programs close.
File Management	Organises files in folders and controls read/write access to storage.
Device Management	Communicates with hardware devices via device drivers.
Security	Manages user accounts, passwords, and access permissions.
User Interface	Provides a GUI or CLI for users to interact with the computer.
Error Handling	Detects and handles hardware or software errors to prevent crashes.
Networking	Manages network connections and data sharing between computers.

5.3 Major Operating Systems

OS	Used On	Key Feature
Microsoft Windows 11	Personal computers, laptops (most common worldwide)	Highly compatible, wide software support
macOS Sonoma	Apple Mac computers and MacBooks	Smooth integration with Apple hardware
Linux (Ubuntu, Fedora)	Servers, developers, advanced users	Free, open-source, highly secure
Android	Smartphones and tablets (most popular mobile OS)	Based on Linux, highly customisable
iOS / iPadOS	Apple iPhones and iPads	Secure, optimised for Apple devices
Chrome OS	Google Chromebooks (cloud-based)	Lightweight, web-focused

■ GUI vs. CLI

GUI (Graphical User Interface): Uses windows, icons, menus, and a mouse. Easy and intuitive to use. Examples: Windows desktop, macOS Finder.

CLI (Command-Line Interface): Uses typed text commands. More powerful for advanced tasks, scripting, and server administration. Examples: Windows Command Prompt, Linux Terminal.

5.4 How the OS Manages Processes

A process is a program that is currently running on the computer. The OS uses a scheduler to decide which process receives CPU time and for how long. This happens so rapidly that multiple programs appear to run simultaneously.

Term	Meaning
Process	An instance of a running program (e.g., one open Word document = one process).
Thread	A smaller unit of a process that can run independently within the same program.
Multitasking	Running multiple processes at the same time (or appearing to do so).
Scheduler	The OS component that decides which process runs next and for how long.
Context Switch	When the OS saves the state of one process and switches to another.

5.5 Memory Management

The OS allocates RAM to programs that need it and releases RAM when programs close. If all physical RAM becomes full, the OS uses virtual memory — a reserved portion of the hard drive that acts as extra RAM, though it is much slower.

Concept	Explanation
RAM Allocation	The OS gives each running program the RAM it needs and monitors usage.
Virtual Memory	A portion of the hard drive (or SSD) used as extra RAM when physical RAM is full. Also called a "page file" on Windows.
Paging	Moving inactive memory blocks from RAM to the hard drive to free up space.
Memory Leak	When a program fails to release RAM it no longer needs — slowing the system.

5.6 File Systems

File System	Used By	Key Feature
NTFS	Windows	Supports large files, security permissions, journaling
FAT32	USB drives, older systems	Compatible with almost all devices
exFAT	USB drives, SD cards	Supports files larger than 4 GB; widely compatible
ext4	Linux	Default Linux file system; reliable and fast
APFS	macOS / iOS	Optimised for SSDs; strong encryption support

5.7 Device Drivers

Device drivers are small programs that allow the OS to communicate with specific hardware devices. Without the correct driver, the device will not work. For example, a printer driver tells the OS exactly how to send data to a particular printer model. Drivers are often installed automatically when a device is connected or when Windows Update runs.

5.8 Utility Programs

Utility	Purpose
Antivirus / Anti-malware	Detects and removes viruses, spyware, and malicious software.
Disk Defragmenter	Rearranges fragmented data on an HDD for faster access.
Disk Cleanup	Removes temporary files and frees up storage space.
Backup Software	Creates copies of important files in case of data loss.
File Compression (ZIP)	Reduces file sizes for easier storage and sharing.
Task Manager	Shows CPU, RAM, and disk usage; lets you close frozen programs.

Utility	Purpose
Firewall	Monitors network traffic and blocks unauthorised access.
System Restore	Returns the OS to a previous working state after a problem.

5.9 User Accounts and Security

Modern operating systems support multiple user accounts on the same device, each with its own files, settings, and level of access.

Account Type	Capabilities
Administrator	Full access to all settings, software installation, and system files.
Standard User	Can use programs and change personal settings but cannot install system-wide software.
Guest	Very limited access; temporary use without permanent settings saved.

■ Best Practice

Use a Standard User account for everyday tasks and only use an Administrator account when installing software or changing system settings. This reduces the risk of malware gaining full system access.

Chapter Review Questions

A. Short Answer Questions

- What is an operating system? Why is it essential?
- List five key functions of an operating system and describe each.
- What is the difference between a GUI and a CLI? Give one example of each.
- What is a device driver? Why is it needed?
- What is virtual memory? When does the OS use it?
- What is the difference between an Administrator account and a Standard User account?
- Name three utility programs and explain what each does.

B. Fill in the Blanks

- The OS manages CPU time between programs through _____ management.
- A _____ interface uses windows, icons, and a mouse.
- _____ is a free, open-source operating system widely used on servers.
- Device _____ allow the OS to communicate with hardware devices.
- _____ software protects the computer from viruses and malware.

C. True or False

- The operating system is an example of application software. (True / False)
- A CLI uses typed commands instead of a graphical interface. (True / False)
- Virtual memory uses part of the hard drive as extra RAM. (True / False)
- A file system determines how data is stored on a drive. (True / False)
- An Administrator account has full access to all system settings. (True / False)

Key Terms Glossary

Term	Definition

Operating System (OS)	System software managing hardware resources and providing a platform for applications.
Process	An instance of a program currently running on the computer.
Multitasking	Running multiple programs or processes at the same time.
GUI	Graphical User Interface — uses windows, icons, and a mouse.
CLI	Command-Line Interface — uses typed text commands.
Virtual Memory	Hard drive space used as extra RAM when physical RAM is full.
Scheduler	The OS component that decides which process runs next and for how long.
File System	A structure determining how data is stored and organised on a storage device.
Device Driver	A program allowing the OS to communicate with a specific hardware device.
Administrator	A user account with full system access and the ability to install software.

— End of Chapter 5 —

CHAPTER 6

FILE MANAGEMENT

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Define files and folders and explain their purpose
- ✓ Identify common file types and their extensions
- ✓ Navigate the file system using File Explorer (Windows) or Finder (macOS)
- ✓ Perform file operations: create, copy, move, rename, and delete
- ✓ Apply good file organisation practices
- ✓ Use search tools to find files quickly
- ✓ View and interpret file properties

6.1 What is a File?

A file is a collection of data stored on a computer under a single name. Every document you write, image you save, song you download, or program you install is stored as a file. Files are identified by their name and extension — the suffix after the dot that tells the computer what type of file it is and which program should open it.

6.2 Common File Types and Extensions

Extension	File Type	Opens With
.docx	Word Document	Microsoft Word, Google Docs
.xlsx	Excel Spreadsheet	Microsoft Excel, Google Sheets
.pptx	PowerPoint Presentation	Microsoft PowerPoint
.pdf	Portable Document Format	Adobe Acrobat, Chrome, PDF Viewer
.txt	Plain Text File	Notepad, any text editor
.jpg / .jpeg	Image (compressed)	Image Viewer, Photoshop
.png	Image (high quality/transparent)	Image Viewer, Photoshop
.mp3	Audio File	VLC, Spotify, Windows Media Player
.mp4	Video File	VLC, Windows Media Player
.zip / .rar	Compressed Archive	WinZip, WinRAR, built-in extractor
.exe	Windows Executable Program	Runs directly (launches/installs)
.py	Python Program	Python interpreter, code editor
.html	Web Page	Any web browser

■ Best Practice

Be careful with unknown .exe, .bat, or .vbs files received by email or downloaded from untrustworthy websites — they may contain malware. Always verify the source before opening executable files.

6.3 Folders and the File System Hierarchy

A folder (also called a directory) is a container that holds files and other folders. The file system is arranged in a hierarchy — like an inverted tree with branches.

- Root: The top of the hierarchy (C:\ on Windows, / on Linux/macOS).
- Folders: Containers that group related files. Can be nested inside each other.
- Subfolders: Folders within folders (e.g., Documents → School → Year 8).
- Files: The actual data stored at the end of the hierarchy.

Example path: C:\Users\Ali\Documents\School\Year8\Notes.docx

6.4 File Explorer and Finder

File Explorer (Windows) and Finder (macOS) are the built-in programs for navigating and managing files and folders. They provide a graphical view of the file system.

Area	Purpose
Navigation Pane (left)	Shows a tree view of drives, folders, and quick-access locations.
Main Area (right)	Displays the contents of the currently selected folder.
Address Bar	Shows the full path of the current location. Click to type a path directly.
Search Box	Type to search for files and folders by name or content.
Ribbon / Toolbar	Contains commands for file operations (copy, move, delete, etc.).
Quick Access	Pinned frequently used folders for fast navigation (e.g., Desktop, Downloads).

6.5 File Operations

Operation	Keyboard Shortcut	Method
Copy	Ctrl + C	Right-click > Copy
Cut (Move)	Ctrl + X	Right-click > Cut
Paste	Ctrl + V	Right-click > Paste
Select All	Ctrl + A	Edit > Select All
Undo	Ctrl + Z	Edit > Undo
Rename	F2	Right-click > Rename
Delete	Delete key	Right-click > Delete

■ Recycle Bin / Trash

Deleted files are sent to the Recycle Bin (Windows) or Trash (macOS) and are NOT permanently deleted yet.

To permanently delete and free up storage space, empty the Recycle Bin.
To recover a deleted file, open the Recycle Bin and click Restore.

6.6 Searching for Files

Method	How to Use
File Explorer Search Box	Type a file name or keyword in the search box at the top-right of File Explorer.
Windows Search (Taskbar)	Press Windows key and type the file name to search across the whole computer.

Method	How to Use
Filters	Use Date modified, File type, or Size filters to narrow search results.
Finder Search (macOS)	Press Cmd + Space to open Spotlight and type your search query.

6.7 File Properties

Every file has metadata — information about the file itself. To view it: right-click the file and select Properties (Windows) or Get Info (macOS).

Property	Description
Name	The file name and extension.
Type	The type of file (e.g., Microsoft Word Document).
Size	How much storage space the file uses.
Location	The full folder path where the file is stored.
Date Created	When the file was first created.
Date Modified	When the file was last saved or changed.
Read-only	If checked, the file cannot be edited without removing this attribute.

6.8 File Organisation Best Practices

- Use clear, descriptive names: "Science_Project_Chapter3.docx" is better than "doc1.docx".
- Create a logical folder structure: sort by subject, project, or year.
- Avoid saving everything to the Desktop — it slows down the computer.
- Delete files you no longer need to free up storage space.
- Back up important files regularly to an external drive or cloud storage.
- Use consistent naming conventions: use dates as YYYY-MM-DD for easy sorting.

6.9 Compressing Files

File compression reduces the size of files by encoding data more efficiently. Compressed archives (ZIP, RAR, 7z) can contain multiple files in a single, smaller package.

Format	Feature
ZIP	Built into Windows and macOS. No extra software needed for basic use.
RAR	Better compression than ZIP. Requires WinRAR to create archives.
7z	Excellent compression. Free and open-source (7-Zip software).

Chapter Review Questions

A. Short Answer Questions

- What is a file? What is a file extension and why is it important?
- What is the difference between a folder and a file?
- What is the difference between copying a file and moving a file?
- What happens when you delete a file in Windows? How do you permanently delete it?
- What is File Explorer? Name four areas of its interface and their purpose.
- How do you search for a file in Windows? Name two methods.
- List five best practices for organising files on a computer.

B. Fill in the Blanks

- A file with the extension .docx is a _____ document.
- The keyboard shortcut to copy a file is _____.
- Deleted files in Windows are sent to the _____ before permanent deletion.
- A _____ is a container that holds files and other folders.
- To view a file's creation date, you right-click and select _____.

C. True or False

- A .exe file is safe to open from any source. (True / False)
- You can recover a file from the Recycle Bin if you have not emptied it. (True / False)
- Ctrl + X cuts (moves) a file to the clipboard. (True / False)
- The root of the file hierarchy on Windows is C:\. (True / False)
- Compressing a file permanently reduces its quality. (True / False)

Key Terms Glossary

Term	Definition
File	A named collection of data stored on a computer.
Folder / Directory	A container that organises files and other folders.
File Extension	The suffix after the dot in a file name (e.g., .docx, .jpg) indicating file type.
File System	The structure used by an OS to store and organise files on a drive.
File Explorer	Windows built-in tool for navigating and managing files and folders.
Path	The full address of a file location (e.g., C:\Users\Ali\Documents\file.txt).
Recycle Bin	Temporary storage for deleted files in Windows before permanent deletion.
Properties	Metadata about a file: name, type, size, creation date, and modification date.
Compression	Reducing file size by encoding data more efficiently (e.g., ZIP format).
Archive	A compressed file containing one or more files or folders (e.g., .zip, .rar).

— End of Chapter 6 —

CHAPTER 7

MS WORD

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Navigate the Microsoft Word interface: Ribbon, tabs, and toolbars
- ✓ Create, open, save, and print documents
- ✓ Format text using font, size, bold, italic, and colour options
- ✓ Apply paragraph formatting: alignment, spacing, and indentation
- ✓ Insert and format tables, images, and headers/footers
- ✓ Use spelling and grammar check to proofread documents
- ✓ Apply page layout settings: margins, orientation, and page size

7.1 What is Microsoft Word?

Microsoft Word is a word processing application that allows users to create, format, and edit text documents. It is one of the most widely used programs in schools, businesses, and homes worldwide. Documents are saved with the .docx extension.

7.2 The Word Interface

Element	Description
Title Bar	Shows the document name and window controls (minimise, maximise, close).
Ribbon	The main toolbar containing all commands, organised into tabs.
Tabs	Categories on the Ribbon: Home, Insert, Layout, References, Review, View.
Quick Access Toolbar	Small toolbar above the Ribbon for Save, Undo, and Redo.
Document Area	The white page where you type and see your content.
Status Bar	Shows page number, word count, and zoom level at the bottom.
Ruler	Horizontal bar showing margins, tabs, and indentation settings.

7.3 Saving a Document

Save Method	Description	Shortcut
Save (Ctrl + S)	Saves changes to the existing file.	Ctrl + S
Save As	Saves a copy with a new name, location, or format.	F12
AutoSave	Available in Microsoft 365 — saves to OneDrive automatically.	Automatic
Export to PDF	File > Export > Create PDF/XPS.	—

■ Best Practice

Save your document every few minutes using Ctrl + S. This prevents losing work if the program or computer crashes unexpectedly.

7.4 Text Formatting

Option	Description	Shortcut
Font Face	Changes the typeface (Arial, Times New Roman, Calibri, etc.).	Font dropdown
Font Size	Changes text size in points — 12pt is standard body text.	Size box
Bold	Makes text thicker and darker.	Ctrl + B
Italic	Slants text to the right.	Ctrl + I
Underline	Adds a line beneath the text.	Ctrl + U
Strikethrough	Draws a line through the text.	Format > Font
Font Colour	Changes the colour of the selected text.	Font Colour button
Superscript / Subscript	Raises or lowers text (x^2 or H_2O).	Ctrl+Shift+= / Ctrl+=

7.5 Paragraph Formatting

Alignment	Description	Shortcut
Align Left	Text lines up at the left margin (default).	Ctrl + L
Centre	Text is centred between the margins.	Ctrl + E
Align Right	Text lines up at the right margin.	Ctrl + R
Justify	Text is spread to fill the full width of the line.	Ctrl + J

7.6 Page Layout

Setting	Options	Where to Find
Page Orientation	Portrait (tall) or Landscape (wide)	Layout tab > Orientation
Page Size	A4, Letter, Legal, and others	Layout tab > Size
Margins	Narrow, Normal, Wide, or Custom values	Layout tab > Margins
Columns	Single, Two, Three, or custom column layout	Layout tab > Columns

7.7 Common Keyboard Shortcuts

Shortcut	Action
Ctrl + N	New document
Ctrl + O	Open document
Ctrl + S	Save
Ctrl + P	Print

Shortcut	Action
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + C / V / X	Copy / Paste / Cut
Ctrl + A	Select all
Ctrl + F	Find
Ctrl + H	Find and Replace
F7	Spelling & Grammar check

7.8 Styles and Headings

A Style is a saved set of formatting settings that can be applied to text in one click. Using Styles keeps your document consistent and automatically builds the document structure. Styles are found in the Home tab > Styles gallery.

Style	Use
Heading 1	Main chapter or section title. Large and prominent.
Heading 2	Sub-section under a Heading 1. Slightly smaller.
Heading 3	Sub-sub-section. Even smaller.
Normal	Standard body text — the default for most paragraphs.
Title	The document title on the first page. Very large.
Subtitle	A secondary title or tagline below the Title style.

Did You Know?

If you apply Heading styles throughout your document, Word can automatically generate a Table of Contents for you — go to References tab > Table of Contents.

7.9 Tables in Word

A table organises information into rows and columns. To insert a table: go to Insert tab > Table, then drag to select the number of rows and columns you need.

Table Operation	How to Do It
Insert a table	Insert tab > Table > drag to select size
Add a row at the bottom	Press Tab in the last cell of the last row
Delete a row or column	Right-click the row/column > Delete
Merge cells	Select cells > right-click > Merge Cells
Change column width	Drag the border between columns
Apply a table style	Click in the table > Table Design tab > choose a style

7.10 Headers and Footers

A header appears at the top of every page. A footer appears at the bottom of every page. They are ideal for page numbers, document titles, dates, or author names.

Inserting a Header or Footer

- Go to Insert tab > Header (or Footer).
- Choose a built-in style or click Edit Header to type your own content.
- Double-click inside the header/footer area to edit it.
- Double-click in the document body to exit the header/footer area.

Inserting Page Numbers

- Insert tab > Page Number > choose a position (Top, Bottom, Margins).
- The page number updates automatically as pages are added or removed.

7.11 Inserting Images and Shapes

Insert Type	Steps
Image from file	Insert tab > Pictures > This Device > choose your image file
Online image	Insert tab > Pictures > Online Pictures > search and insert
Shape	Insert tab > Shapes > click a shape > draw it on the page
Text Box	Insert tab > Text Box > draw it > type inside
SmartArt	Insert tab > SmartArt > choose a diagram type

7.12 Spelling and Grammar Check

Word automatically underlines potential errors: red for spelling, blue or green for grammar. You can also run a full check at any time.

- Press F7 or go to Review tab > Spelling & Grammar to start the check.
- Right-click an underlined word to see suggested corrections.
- Use the Thesaurus (Review tab) to find synonyms for a word.
- Word Count: Review tab > Word Count — shows words, characters, and pages.

■ Best Practice

Always proofread your document manually even after using spell check. Spell check will not catch correctly spelled but wrong words — for example "their" instead of "there".

Chapter Review Questions

A. Short Answer Questions

- What is Microsoft Word? What file extension do Word documents use?
- What is the Ribbon? Name four tabs found on it.
- What is the difference between Save and Save As?
- Explain three text formatting options with their keyboard shortcuts.
- What is a Style in Word? Why should you use Heading styles?
- What is the difference between a header and a footer? How do you insert page numbers?
- How do you insert a table in Word? Name three operations you can do on a table.
- Why should you not rely solely on spell check to proofread a document?

B. Fill in the Blanks

- The keyboard shortcut to save a document is _____.
- To centre text, press Ctrl + _____.
- Word documents use the _____ file extension.
- The _____ tab contains formatting options like Bold and Italic.

- Pressing F7 opens the _____ and Grammar check.
- A _____ is a saved set of formatting settings applied in one click.

C. True or False

- Ctrl + B makes selected text italic. (True / False)
- Headers appear at the top of every page in a document. (True / False)
- Ctrl + Z undoes the last action. (True / False)
- Page orientation options are found on the Home tab. (True / False)
- Word's spell checker will catch all errors including "their" vs "there". (True / False)

Key Terms Glossary

Term	Definition
Word Processor	Software for creating, editing, and formatting text documents.
Ribbon	The main toolbar in Word, organised into tabs.
Font	The typeface used for text (e.g., Arial, Times New Roman).
Formatting	Changing the appearance of text or paragraphs.
Header / Footer	Text appearing at the top / bottom of every page.
Margin	The blank space between text and the edge of the page.

— End of Chapter 7 —

CHAPTER 8

MS POWERPOINT

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Describe the PowerPoint interface and its key components
- ✓ Create, add, delete, and organise slides
- ✓ Apply slide layouts, themes, and design options
- ✓ Insert and format text, images, shapes, and SmartArt
- ✓ Add slide transitions and object animations
- ✓ Run and navigate a slide show presentation

8.1 What is Microsoft PowerPoint?

Microsoft PowerPoint is a presentation software application used to create visual slideshows. Presentations are made up of individual slides containing text, images, charts, shapes, and other media. PowerPoint files are saved with the .pptx extension.

8.2 The PowerPoint Interface

Element	Description
Slide Panel (left)	Thumbnails of all slides. Click one to navigate to that slide.
Editing Area (centre)	The main workspace where you design the current slide.
Notes Panel (bottom)	Area for typing speaker notes visible only to the presenter.
Ribbon	All commands in tabs: Home, Insert, Design, Transitions, Animations, Slide Show.
View Buttons	Switch between Normal, Slide Sorter, Notes Page, and Reading view.

8.3 Slide Layouts

Layout	Best For
Title Slide	Opening slide with a large title and subtitle.
Title and Content	Title at top, content area below — the most common layout.
Two Content	Two side-by-side content areas for comparisons.
Blank	Empty slide — place any content anywhere freely.
Picture with Caption	Large image with a descriptive caption below.
Section Header	Dividers between major sections of a long presentation.

8.4 Themes and Design

A theme applies a consistent set of colours, fonts, and background styles to your entire presentation. Find themes on the Design tab.

■ Best Practice

Use no more than 3 fonts per presentation. Keep text on slides brief — use bullet points, not full paragraphs. Ensure high contrast between text and background for readability.

8.5 Slide Transitions and Animations

Feature	What It Does	Where to Find
Transition	Effect when moving between slides.	Transitions tab
Animation	Effect on individual objects on a slide.	Animations tab
Duration	How long the effect takes to complete.	Both tabs
Apply To All	Apply same transition to every slide.	Transitions tab

8.6 Running a Presentation

Action	How
Start from beginning	F5 or Slide Show tab > From Beginning
Start from current slide	Shift + F5
Next slide	Click, Enter, or right arrow key
Previous slide	Backspace or left arrow
End / exit	Press Escape

8.7 Working with Images and Shapes

Visual elements make presentations more engaging and memorable. PowerPoint supports images, shapes, icons, and SmartArt diagrams.

Element	How to Insert	Best Use
Image	Insert > Pictures > This Device or Online Pictures	Photographs, logos, illustrations
Shape	Insert > Shapes > draw on slide	Arrows, callouts, rectangles for labels
Icon	Insert > Icons > search and insert	Simple vector symbols (free to resize)
Text Box	Insert > Text Box > draw and type	Adding text anywhere outside placeholders
Screenshot	Insert > Screenshot > select window	Capturing software or website screens

Formatting Images

- Click an image to select it, then use the Picture Format tab.
- Drag corner handles to resize proportionally (hold Shift to keep ratio).
- Use Remove Background to cut out image backgrounds automatically.
- Apply Picture Styles (frames, shadows, reflections) from the Format tab.

8.8 SmartArt — Professional Diagrams

SmartArt automatically creates polished diagrams from a list of text. It is perfect for showing processes, hierarchies, cycles, and relationships.

SmartArt Type	Best For
List	Showing non-sequential items or bullet points visually.
Process	A series of steps from start to finish (e.g., a workflow).
Cycle	A repeating circular process (e.g., water cycle, feedback loop).
Hierarchy	Organisational charts — who reports to whom.
Relationship	Showing connections between ideas (Venn diagrams, matrices).
Matrix	Comparing items across two dimensions.

Insert tab > SmartArt > choose a type > type your text in the left panel. PowerPoint arranges it into a polished diagram automatically.

8.9 Presenter View

Presenter View lets you see your speaker notes, a slide preview, and a timer on your own screen — while the audience only sees the current slide on the projector.

- Slide Show tab > check Use Presenter View.
- Your current slide shows on the projector. Your laptop shows: the current slide, the next slide preview, your speaker notes, and a timer.
- Use it to stay on track without reading from the slides in front of the audience.

8.10 Slide Master

The Slide Master controls the design template for all slides. If you want every slide to share the same logo, font, or background, edit the Slide Master instead of changing each slide individually.

- View tab > Slide Master to open it.
- Changes made in the Slide Master apply automatically to all slides.
- Close the Slide Master view when done: Slide Master tab > Close Master View.

Chapter Review Questions

A. Short Answer Questions

- What is Microsoft PowerPoint? What file extension does it use?
- Name four elements of the PowerPoint interface and their purpose.
- What is a slide layout? Name three layouts and when to use each.
- What is the difference between a transition and an animation?
- What is SmartArt? Name three SmartArt types and what each is best for.
- What is Presenter View? What advantages does it give the presenter?
- What is the Slide Master and when would you use it?

B. Fill in the Blanks

- PowerPoint files are saved with the _____ extension.
- The _____ tab contains slide themes and backgrounds.
- Pressing F5 starts the slideshow from the _____.
- Speaker notes are typed in the _____ panel.
- The _____ controls the design template for all slides at once.

C. True or False

- A transition plays when moving from one slide to the next. (True / False)
- SmartArt creates professional diagrams automatically from text. (True / False)
- You should use as many different fonts as possible in a presentation. (True / False)

- Presenter View shows your notes and a timer while the audience sees only the slide. (True / False)

Key Terms Glossary

Term	Definition
Presentation	A set of slides used to communicate ideas visually to an audience.
Slide	A single page within a PowerPoint presentation.
Layout	A predefined arrangement of placeholders on a slide.
Theme	Coordinated colours, fonts, and backgrounds applied to all slides.
Transition	An animation effect that plays when moving between slides.
Animation	An effect applied to an individual object on a slide.
Placeholder	A designated area on a slide for content (text, image, media).

— End of Chapter 8 —

CHAPTER 9

MS EXCEL

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Describe the Excel interface and understand rows, columns, and cells
- ✓ Enter and format data in a spreadsheet
- ✓ Write basic formulas using arithmetic operators
- ✓ Use common functions: SUM, AVERAGE, MAX, MIN, COUNT, IF
- ✓ Understand relative and absolute cell references
- ✓ Create and format charts from spreadsheet data
- ✓ Sort and filter data in Excel

9.1 What is Microsoft Excel?

Microsoft Excel is a spreadsheet application used to store, organise, and analyse numerical data. Spreadsheets are made up of cells arranged in numbered rows and lettered columns. Excel files use the .xlsx extension.

9.2 The Excel Interface

Element	Description
Cell	The intersection of a row and column (e.g., B3 = column B, row 3).
Cell Reference	The unique address of a cell (e.g., A1, B3, C10).
Formula Bar	Shows the content or formula in the selected cell.
Name Box	Displays the reference of the currently selected cell.
Sheet Tabs	Each tab is a separate worksheet within the workbook.
Ribbon	All commands in tabs: Home, Insert, Formulas, Data, Review.

9.3 Formulas and Functions

A formula is an instruction that tells Excel to calculate a result. All formulas begin with the = sign.

Function	Syntax	What It Does
SUM	=SUM(A1:A10)	Adds all values in the range.
AVERAGE	=AVERAGE(A1:A10)	Calculates the mean of the range.
MAX	=MAX(A1:A10)	Returns the largest value.
MIN	=MIN(A1:A10)	Returns the smallest value.
COUNT	=COUNT(A1:A10)	Counts cells containing numbers.
IF	=IF(B2>=50,"Pass","Fail")	Returns one value if TRUE, another if FALSE.

9.4 Cell References

Reference Type	Example	Behaviour When Copied
Relative	=A1	Row and column both adjust.
Absolute	=$A$1	Neither row nor column changes.
Mixed (col fixed)	=$A1	Column stays; row adjusts.
Mixed (row fixed)	=A$1	Row stays; column adjusts.

9.5 Charts

Chart Type	Best For
Bar / Column Chart	Comparing values across categories (e.g., sales by month).
Line Chart	Showing trends over time (e.g., temperature changes).
Pie Chart	Showing proportions as parts of a whole (e.g., budget breakdown).
Scatter Plot	Showing relationships between two variables.

9.6 Common Keyboard Shortcuts

Shortcut	Action
Ctrl + S	Save the workbook
Ctrl + Home	Go to cell A1
F2	Edit the selected cell
Delete	Clear cell content
Alt + =	AutoSum (insert SUM formula)
Ctrl + 1	Open Format Cells dialogue

9.7 Sorting and Filtering

Sorting rearranges your data in a specific order. Filtering hides rows that do not match a condition, letting you focus on what you need.

Sorting Data

- Select any cell in your data range.
- Go to Data tab > Sort A to Z (ascending) or Sort Z to A (descending).
- For sorting by multiple columns: Data tab > Sort > Add Level.

Filtering Data

- Select any cell in the header row of your table.
- Data tab > Filter — dropdown arrows appear on each column header.
- Click a dropdown arrow > choose values to show or hide.
- To remove the filter: Data tab > Filter again, or clear individual filters.

9.8 Conditional Formatting

Conditional Formatting automatically highlights cells that meet a condition — for example, all values above 90 turn green, or all failed scores turn red.

Rule Type	Example Use
Highlight Cell Rules > Greater Than	Highlight scores above 80 in green.
Highlight Cell Rules > Less Than	Highlight scores below 50 in red.
Data Bars	Show bars inside cells — longer = larger value.
Colour Scales	Heat-map colouring — green for high, red for low.
Icon Sets	Add arrows or traffic lights next to values.
New Rule	Custom formula-based rule for complex conditions.

How to apply: Select your data range > Home tab > Conditional Formatting > choose a rule type.

9.9 Freeze Panes

When you have a large spreadsheet, the header row disappears as you scroll down. Freeze Panes keeps row or column headers visible at all times.

- Click the cell below the row and to the right of the column you want to freeze.
- View tab > Freeze Panes > Freeze Panes.
- To freeze just the top row: View tab > Freeze Panes > Freeze Top Row.
- To unfreeze: View tab > Freeze Panes > Unfreeze Panes.

9.10 VLOOKUP — Finding Data

VLOOKUP (Vertical Lookup) searches for a value in the first column of a table and returns a related value from another column in the same row.

Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

Argument	Meaning	Example
lookup_value	The value you are searching for.	A2 (a student ID)
table_array	The data table to search in.	$B$2:$D$100
col_index_num	Which column of the table to return.	3 (returns column 3)
range_lookup	FALSE = exact match (almost always use FALSE).	FALSE

Example: =VLOOKUP(A2,$B$2:$D$100,3,FALSE) — looks for the value in A2 within the first column of the range B2:D100 and returns what is in column 3 of that range.

Chapter Review Questions

A. Short Answer Questions

- What is a spreadsheet? What is Microsoft Excel used for?
- What is a cell reference? Give two examples.
- What is the difference between a formula and a function in Excel?
- Explain the difference between a relative and an absolute cell reference.
- Name four Excel functions and explain each.
- What is Conditional Formatting? Give an example of when you would use it.
- What is Freeze Panes? When would it be useful?
- What does VLOOKUP do? Write an example with the correct syntax.

B. Fill in the Blanks

- All Excel formulas begin with the _____ sign.

- The function _____ adds all values in a selected range.
- An _____ reference uses $ symbols to prevent row/column changes when copying.
- Excel files use the _____ extension.
- _____ Formatting automatically highlights cells that meet a condition.

C. True or False

- A relative cell reference adjusts when a formula is copied. (True / False)
- =MAX(A1:A10) returns the smallest value. (True / False)
- A bar chart is best for showing proportions as parts of a whole. (True / False)
- The formula bar shows the content of the currently selected cell. (True / False)
- Freeze Panes keeps header rows visible as you scroll down a large spreadsheet. (True / False)

Key Terms Glossary

Term	Definition
Spreadsheet	A document of rows and columns for storing and analysing data.
Cell	The intersection of a row and column — the basic data unit.
Formula	An instruction starting with = that tells Excel to calculate.
Function	A built-in formula: SUM, AVERAGE, MAX, MIN, COUNT, IF.
Relative Reference	A cell reference that adjusts when a formula is copied.
Absolute Reference	A cell reference locked with $ that does not change when copied.
Chart	A graphical representation of spreadsheet data.

— End of Chapter 9 —

CHAPTER 10

INTERNET BASICS

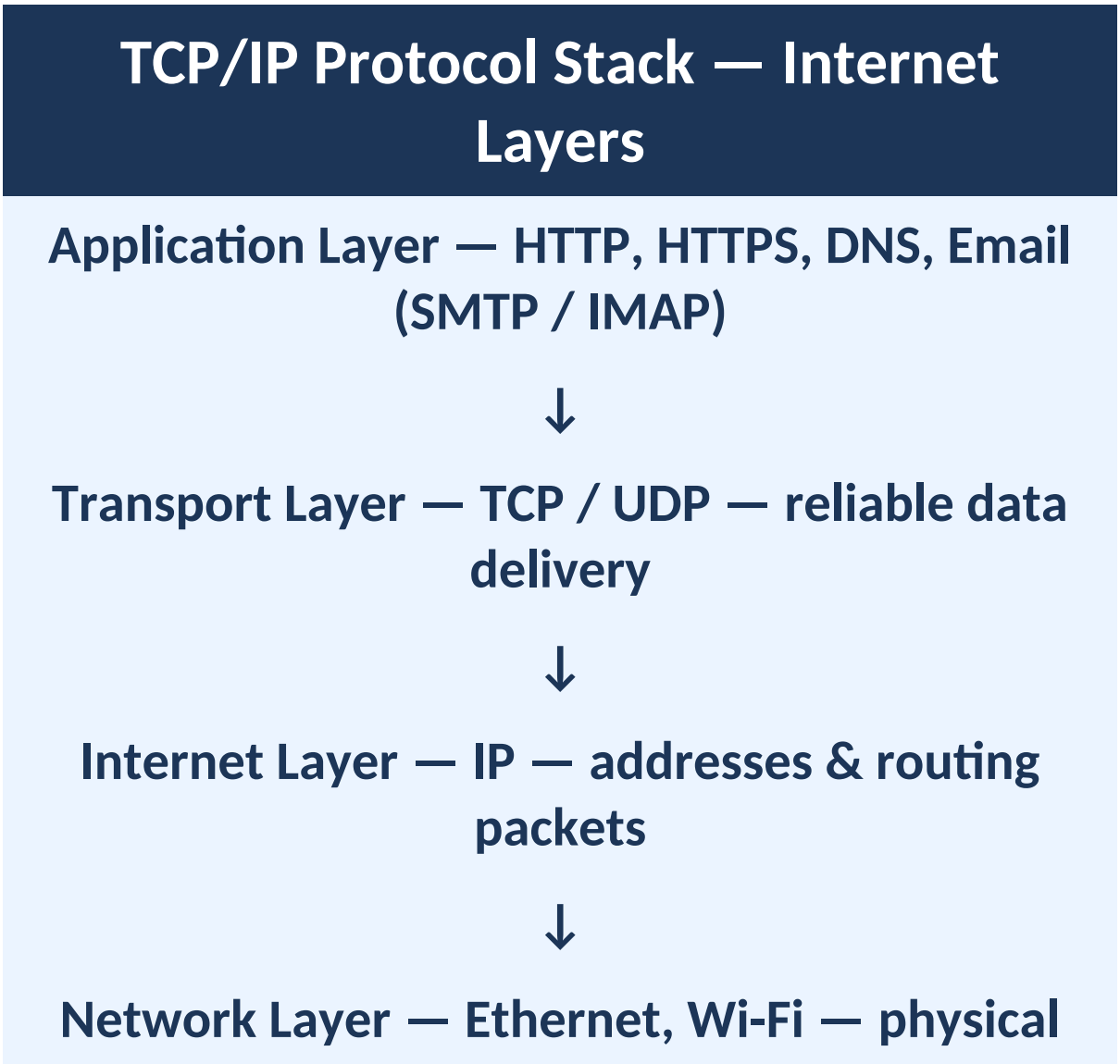
Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Explain what the internet is and how it works
- ✓ Distinguish between the Internet and the World Wide Web
- ✓ Describe key internet services: web, email, cloud, streaming
- ✓ Understand IP addresses, DNS, and how browsers work
- ✓ Identify safe and unsafe online practices

10.1 What is the Internet?

The Internet is a global network of billions of computers and devices all connected together using standardised communication protocols. It allows users worldwide to communicate, share information, and access services from anywhere.



transmission

Figure 10.1: The TCP/IP stack — layers that make internet communication possible

10.2 How the Internet Works

Data on the internet is broken into small units called packets. Each packet travels independently and is reassembled at the destination. This is called packet switching.

Concept	Description
IP Address	A unique numerical address for every device (e.g., 192.168.1.1 for IPv4).
DNS	Domain Name System — translates domain names (google.com) into IP addresses.
Router	A device that forwards data packets between networks.
ISP	Internet Service Provider — the company providing your connection.
Bandwidth	Maximum data transmitted per second. Measured in Mbps or Gbps.
HTTP/HTTPS	Protocols for web pages. HTTPS is the secure, encrypted version.

■ Internet vs. World Wide Web

The Internet is the physical infrastructure — cables, routers, servers connecting billions of devices.

The World Wide Web is just one service running on the internet — websites accessed via a browser.

Email, gaming, and streaming also use the internet but are NOT the Web.

10.3 Internet Services

Service	Description	Examples
World Wide Web	Web pages accessed via a browser using HTTP/HTTPS.	Google, Wikipedia, YouTube
Email	Electronic messages sent and received over the internet.	Gmail, Outlook, Yahoo Mail
Cloud Computing	Using remote servers to store, manage, and process data.	Google Drive, OneDrive, AWS
Streaming	Continuous delivery of audio or video over the internet.	Netflix, Spotify, YouTube
VoIP	Phone calls via the internet.	WhatsApp, Zoom, Skype
Social Media	Platforms for sharing and interacting with content.	Instagram, Twitter/X, TikTok

10.4 Browsers and Search Engines

Type	Examples
Web Browsers	Google Chrome, Mozilla Firefox, Microsoft Edge, Apple Safari
Search Engines	Google, Bing, DuckDuckGo, Yahoo Search

10.5 Types of Internet Connection

There are several technologies used to deliver internet access to homes, schools, and businesses. They differ in speed, cost, and availability.

Connection Type	Technology	Typical Speed	Notes
Fibre Optic (FTTH)	Light through glass cables	100 Mbps – 1 Gbps+	Fastest available; ideal for homes and schools
ADSL Broadband	Copper phone lines	5 – 20 Mbps	Common but slower; speed drops with distance
Cable Broadband	TV cable network	50 – 300 Mbps	Fast; shared with neighbours
4G Mobile Data	Cellular towers	10 – 100 Mbps	Wireless; used in smartphones and mobile routers
5G Mobile Data	Next-gen cellular	100 Mbps – 10 Gbps	Very fast; still being rolled out in many areas
Wi-Fi	Wireless LAN (radio)	Varies by router	Connects local devices to a router wirelessly
Satellite Internet	Orbiting satellites	25 – 500 Mbps	Works anywhere; higher latency

10.6 Web Security and Privacy

Every time you browse the web, websites and your browser collect data. Understanding how this works helps you protect your privacy online.

HTTPS and SSL/TLS

When a website uses HTTPS (instead of HTTP), data between your browser and the website is encrypted using SSL/TLS. Look for the padlock icon in the address bar — never enter passwords or payment details on a site without it.

Cookies

Cookie Type	Purpose	Example
Session cookies	Keep you logged in during a browsing session. Deleted when you close the browser.	Shopping cart items
Persistent cookies	Remember your preferences across sessions.	Saved username, language setting
Third-party cookies	Placed by advertisers to track you across multiple websites.	Targeted advertising

Protecting Your Privacy Online

- Use a browser with built-in privacy features (Firefox, Brave, DuckDuckGo).
- Clear your browsing history and cookies regularly.
- Use Incognito/Private mode for sensitive browsing — cookies are deleted when you close the window.
- Use a VPN (Virtual Private Network) to encrypt your traffic and hide your IP address.
- Review and limit app permissions — social media apps should not access your location or contacts unnecessarily.

10.7 Downloading and Uploading

Term	Meaning	Example
Download	Receiving data from the internet to your	Watching a video, saving a PDF

Term	Meaning	Example
	device.	
Upload	Sending data from your device to the internet.	Posting a photo, submitting coursework
Download speed	How fast data arrives at your device (Mbps).	Affects streaming, web browsing
Upload speed	How fast data leaves your device (Mbps).	Affects video calls, file sharing
Latency (ping)	Delay between sending and receiving data (ms).	Affects online gaming and video calls

Chapter Review Questions

A. Short Answer Questions

- What is the internet? How does packet switching work?
- What is DNS, and why is it useful? Give an example.
- What is the difference between the Internet and the World Wide Web?
- Name four internet services and briefly describe each.
- What is an IP address? What does a URL consist of?

B. Fill in the Blanks

- The _____ translates domain names into IP addresses.
- HTTP stands for Hypertext Transfer _____.
- A _____ forwards data packets between networks.
- The Internet began as a US military project called _____.

C. True or False

- The World Wide Web is the same as the Internet. (True / False)
- An IP address uniquely identifies a device on the internet. (True / False)
- Google is a web browser. (True / False)
- HTTPS is the secure version of HTTP. (True / False)

Key Terms Glossary

Term	Definition
Internet	A global network connecting billions of computers using standard protocols.
World Wide Web (WWW)	A system of websites accessible through a browser.
IP Address	A unique numerical address identifying a device on a network.
DNS	Domain Name System — converts domain names to IP addresses.
Bandwidth	Maximum data transfer speed (Mbps or Gbps).
Browser	Software for accessing and displaying web pages.
URL	Uniform Resource Locator — the complete web address of a page.
Packet Switching	Breaking data into packets that travel independently and are reassembled.

— End of Chapter 10 —

CHAPTER 11

EMAIL AND ONLINE COMMUNICATION

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Explain what email is and describe the structure of an email message
- ✓ Compose, send, reply to, and forward emails correctly
- ✓ Understand the difference between To, CC, and BCC fields
- ✓ Apply professional email etiquette
- ✓ Describe other communication tools: messaging, video calls, forums
- ✓ Practise safe and responsible online communication

11.1 What is Email?

Email (electronic mail) is a digital method of exchanging messages over the internet. It allows users to send text, attachments, and formatted content to one or more recipients almost instantly, anywhere in the world.

11.2 Email Address Structure

Every email user has a unique email address in the format: username@domain.extension

Part	Description	Example
Username	Chosen name identifying the user account.	ali.hassan
@ Symbol	Separates the username from the domain.	@
Domain	The organisation or email service provider.	gmail.com, school.edu
Extension / TLD	Top-Level Domain — type or country.	.com, .org, .edu, .gov

11.3 Structure of an Email

Field	Purpose
From	Sender's email address — filled automatically.
To	Main recipient(s). Required.
CC (Carbon Copy)	Additional recipients who get a copy. All can see who is CC'd.
BCC (Blind Carbon Copy)	Recipients who get a copy but are hidden from all others.
Subject	Brief title describing the purpose. Always include a clear subject.
Body	The main content of the message.
Signature	Closing section: name, title, contact info.
Attachment	Files added to the email. Size limits apply.

■ CC vs. BCC

Use CC when all recipients should know who else received the email.

Use BCC to send to many people without revealing each address to others — useful for newsletters and group announcements.

11.4 Email Etiquette

Good Practice	Poor Practice
Write a clear, informative subject line.	Leaving the subject line blank.
Use proper grammar and punctuation.	Writing with typos or text-speak.
Be concise and clear.	Writing unnecessarily long emails.
Use a professional greeting and sign-off.	Starting with just "Hey" in formal contexts.
Reply within 24–48 hours.	Leaving important emails unanswered for days.
Use BCC when emailing large groups.	CC'ing everyone and exposing all addresses.

11.5 Other Online Communication Tools

Tool	Type	Examples	Best For
Instant Messaging	Real-time text chat	WhatsApp, Telegram, iMessage	Quick informal conversations
Video Calling	Live video and audio	Zoom, Google Meet, Teams	Remote meetings and lessons
Discussion Forums	Threaded discussions	Reddit, Stack Overflow	Q&A, sharing knowledge
Social Media	Content sharing	Instagram, TikTok, Facebook	Broadcasting and networking

11.6 Managing Your Inbox

A well-organised inbox makes it faster to find important emails and reduces stress. Most email providers offer powerful tools for organising messages.

Feature	Purpose	How to Use
Folders / Labels	Group related emails together.	Create folders (e.g., School, Work) and move emails into them.
Filters / Rules	Automatically sort incoming emails.	Set up rules: emails from a certain sender go to a specific folder.
Archive	Store emails without deleting them.	Press Archive — email disappears from inbox but is searchable.
Flag / Star	Mark important emails to act on later.	Click the flag or star icon next to an email.
Search	Find specific emails instantly.	Type a keyword, sender, or date in the search bar.
Unsubscribe	Stop receiving unwanted newsletters.	Click Unsubscribe at the bottom of marketing emails.

11.7 Recognising Unsafe Emails

Phishing emails are designed to trick you into clicking malicious links or revealing passwords. Learning to spot them is an essential digital skill.

Warning Sign	Description
Unknown sender	The email address looks strange or is from a domain you do not recognise.
Urgent language	"Act now!", "Your account will be closed!", "Last warning!" — pressure tactics.
Generic greeting	"Dear Customer" or "Dear User" instead of your actual name.
Suspicious links	Hover over links before clicking — the real URL often does not match the text shown.
Poor spelling/grammar	Legitimate companies proofread their emails carefully.
Unexpected attachment	An attachment you were not expecting — especially .exe, .zip, or .docx files.
Requests for personal data	Legitimate organisations never ask for passwords or PINs by email.

■ Best Practice

When in doubt, do not click. If an email claims to be from your bank or school and asks for urgent action — go directly to their official website by typing the address yourself, or call them. Never click the link in the suspicious email.

Chapter Review Questions

A. Short Answer Questions

- What is email? What is the structure of an email address?
- What is the difference between CC and BCC? When should you use each?
- List five rules of good email etiquette.
- Name four types of online communication tools and describe each.
- Name three features you can use to organise your inbox. Explain each.
- List five warning signs that an email may be a phishing attempt.

B. Fill in the Blanks

- The _____ field shows recipients who are hidden from all others.
- Writing in ALL CAPITALS in email is considered _____.
- The subject line should be clear and _____.
- Reply _____ sends your response only to the original sender.
- Emails archived in Gmail _____ from the inbox but remain searchable.

C. True or False

- You should reply to professional emails within 24–48 hours. (True / False)
- BCC recipients can see all other addresses in the email. (True / False)
- You should open attachments from unknown senders immediately. (True / False)
- Legitimate organisations never ask for passwords by email. (True / False)
- Using filters in your email client can automatically sort incoming messages. (True / False)

Key Terms Glossary

Term	Definition
Email	Electronic mail — digital messages sent over the internet.
CC	Carbon Copy — sends a visible copy to additional recipients.
BCC	Blind Carbon Copy — sends a hidden copy; recipients invisible to others.

Attachment	A file (document, image) added to an email message.
Netiquette	Accepted rules of polite, professional digital communication.
Spam	Unsolicited bulk email, often promotional or malicious.
Phishing	A fraudulent email designed to steal passwords or personal data.

— End of Chapter 11 —

CHAPTER 12

CYBER SAFETY AND DIGITAL ETIQUETTE

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Identify common cyber threats: malware, phishing, hacking, ransomware
- ✓ Understand how to protect personal information online
- ✓ Create strong passwords and understand two-factor authentication
- ✓ Define and practise good digital etiquette
- ✓ Know what to do if you experience cyberbullying or an online threat

12.1 Why Cyber Safety Matters

The internet offers tremendous benefits, but it also carries risks. Understanding these risks and how to protect yourself is a vital skill in the modern world. Cyber threats can affect individuals, schools, businesses, and even governments.

12.2 Common Cyber Threats

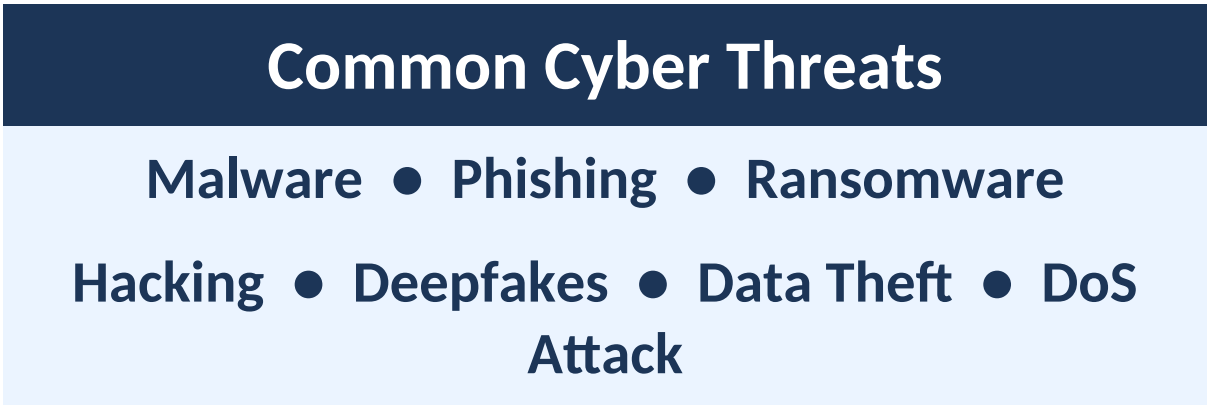


Figure 12.1: Common cyber threats — every connected device is potentially at risk

Threat	Description	Example
Malware	Malicious software designed to damage or gain unauthorised access.	Viruses, worms, spyware, trojans
Phishing	Fake emails or websites tricking users into revealing passwords.	"Your account has been locked — click here to verify."
Ransomware	Malware that encrypts files and demands payment to restore access.	WannaCry attack (2017)
Hacking	Unauthorised access to computer systems or accounts.	Breaking into someone's email account
Deepfakes	AI-generated fake images or videos designed to deceive.	Fake video of a politician
Data Theft	Stealing personal or financial data for fraud.	Stealing credit card numbers
DoS Attack	Flooding a server with requests to make it unavailable.	Taking down a website with traffic

12.3 Protecting Yourself Online

- Use strong, unique passwords for every account (12+ characters, mix of letters, numbers, symbols).
- Enable Two-Factor Authentication (2FA) on all important accounts.
- Keep your operating system and software updated — updates patch security vulnerabilities.
- Use a reputable antivirus program and keep it current.
- Never click suspicious links or download attachments from unknown senders.
- Be cautious about personal information you share on social media.
- Use HTTPS websites only when entering personal or financial information.
- Back up important data regularly to an external drive or cloud storage.

12.4 Strong Passwords

Weak Password ✗	Strong Password ✓	Why?
password123	Tr0ub4dor&3!xQ	Mix of cases, numbers, symbols — hard to guess
123456	B@n@n@-Sunr1se!2024	Long phrase with substitutions — memorable yet secure
john1990	P#7!mK$vLq@2Zr	Completely random — ideal for a password manager

■ Two-Factor Authentication (2FA)

2FA adds a second layer of security beyond your password. After entering your password, you must also provide a one-time code sent by SMS or generated by an app.

Even if someone steals your password, they cannot log in without the second factor.

12.5 Cyberbullying

Cyberbullying is the use of technology to repeatedly harass, humiliate, or harm another person. It can occur through social media, messaging apps, gaming platforms, or email.

If You Experience Cyberbullying

- Do not respond to the bully — responding often encourages them.
- Take screenshots as evidence before blocking.
- Block and report the bully on the platform.
- Tell a trusted adult: parent, teacher, or school counsellor.

12.6 Digital Etiquette

Digital etiquette (or "netiquette") refers to the respectful, responsible, and ethical use of technology and online communication.

- Treat others online with the same respect you would show in person.
- Think before you post — once something is online, it may be permanent.
- Do not share other people's personal information without permission.
- Avoid using all capital letters — it comes across as shouting.
- Credit the original creator when sharing others' work.
- Verify information before sharing — do not spread misinformation.

Chapter Review Questions

A. Short Answer Questions

- What is malware? Name three types and explain each.
- What is phishing? Give an example of a phishing message.
- What is ransomware? Why is it particularly dangerous?
- What is two-factor authentication? Why is it important?
- List five practices for staying safe online.
- What is cyberbullying? What should you do if you experience it?
- What is digital etiquette? Give three rules of good netiquette.

B. Fill in the Blanks

- Malicious software designed to damage a computer is called _____.
- _____ emails trick users into revealing passwords or personal information.
- _____ Authentication requires a password plus a second verification step.
- A strong password should be at least _____ characters long.
- Always look for _____ in the address bar before entering payment details.

C. True or False

- A strong password should be at least 12 characters long. (True / False)
- Two-Factor Authentication makes accounts less secure. (True / False)
- It is safe to click on links from unknown senders. (True / False)
- HTTPS indicates a website uses an encrypted connection. (True / False)
- Using the same password for all accounts is good practice. (True / False)

Key Terms Glossary

Term	Definition
Malware	Malicious software designed to damage or gain unauthorised access.
Phishing	Fraudulent attempt to obtain passwords through fake emails/websites.
Ransomware	Malware that encrypts files and demands payment for access.
Hacking	Gaining unauthorised access to computer systems or accounts.
Two-Factor Authentication	Security requiring a password plus a second verification step.
Cyberbullying	Using technology to repeatedly harass or harm another person.
Deepfake	An AI-generated realistic fake image or video.
Netiquette	Rules of polite, respectful behaviour in digital communication.

— End of Chapter 12 —

CHAPTER 13

LOGICAL THINKING AND PROBLEM SOLVING

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Define computational thinking and explain its four pillars
- ✓ Apply decomposition to break problems into smaller parts
- ✓ Recognise patterns and use them to solve problems efficiently
- ✓ Apply abstraction to focus on essential details
- ✓ Design step-by-step algorithms to solve problems
- ✓ Apply computational thinking to real-world problems

13.1 What is Computational Thinking?

Computational thinking is a problem-solving approach that involves breaking down complex problems and expressing solutions in a way that both humans and computers can understand. It is not just for programmers — it is a fundamental skill for logical reasoning in any context.

Pillar	Key Idea
Decomposition	Break a big problem into smaller, manageable parts.
Pattern Recognition	Find similarities, trends, and repeating structures.
Abstraction	Focus on important details; ignore what is not needed.
Algorithm Design	Create a step-by-step solution to the problem.

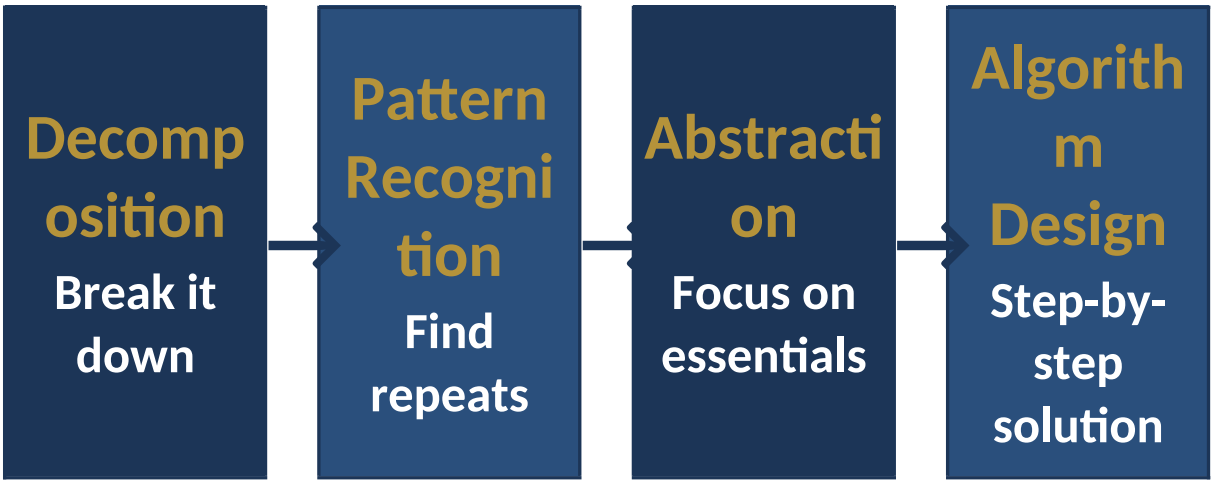


Figure 13.1: The four pillars of computational thinking

13.2 Decomposition

Decomposition means breaking a large, complex problem into smaller, simpler sub-problems. Each sub-problem is easier to understand, solve, and check individually.

Real-world example — Organising a School Event:

- Venue and seating arrangement.
- Invitations and communication.

- Programme and schedule.
- Catering and refreshments.
- Audio-visual equipment.

■ Decomposition in Computing

When a programmer writes a large application, they break it into functions, modules, and classes — each solving one specific sub-problem. This makes code easier to write, test, and maintain.

13.3 Pattern Recognition

Pattern recognition means identifying similarities, trends, repetitions, or regularities in a problem or dataset. Recognising patterns helps us solve new problems faster and write more efficient code.

- 2, 4, 6, 8, 10 ... (adding 2 each time — arithmetic sequence)
- 1, 2, 4, 8, 16 ... (doubling each time — geometric sequence)

In programming: if the same action repeats many times, that is a pattern — solve it with a loop or a function.

13.4 Abstraction

Abstraction means focusing only on the important, relevant details of a problem and ignoring everything that is not needed.

Real-World Object	Key Abstractions Used	Irrelevant Details Ignored
A Map	Roads, distances, landmarks, directions.	Actual terrain, building materials, colours.
A Recipe	Ingredients, quantities, steps in order.	The chef's mood, kitchen brand, bowl colour.
A Computer	Input, Process, Output (IPO model).	Billions of transistors firing inside the CPU.

13.5 Algorithm Design

An algorithm is a precise, ordered sequence of steps to solve a problem. It is the fourth pillar of computational thinking.

- Correctness: Produces the right output for all valid inputs.
- Clarity: Each step is unambiguous and easy to understand.
- Finiteness: The algorithm terminates after a finite number of steps.
- Efficiency: Uses as little time and memory as possible.
- Generality: Works for all valid inputs, not just specific cases.

Chapter Review Questions

A. Short Answer Questions

- What is computational thinking? Why is it important beyond programming?
- Explain decomposition with a real-world example.
- What is pattern recognition? How does it help in programming?
- Explain abstraction and give one everyday example.
- What are the five properties of a good algorithm?

B. Fill in the Blanks

- _____ means breaking a large problem into smaller, manageable parts.
- Identifying similarities and trends in data is called _____ recognition.
- _____ means focusing on important details while ignoring irrelevant ones.
- A step-by-step set of instructions for solving a problem is called an _____.

C. True or False

- Computational thinking is a skill only programmers need. (True / False)
- Abstraction means including every possible detail. (True / False)
- Decomposition breaks a large problem into smaller, manageable parts. (True / False)
- A good algorithm must terminate after a finite number of steps. (True / False)

Key Terms Glossary

Term	Definition
Computational Thinking	A problem-solving approach: decomposition, patterns, abstraction, algorithms.
Decomposition	Breaking a complex problem into smaller, manageable sub-problems.
Pattern Recognition	Identifying similarities, trends, or repeated structures.
Abstraction	Focusing on essential information while ignoring irrelevant details.
Algorithm	A precise, ordered set of steps to solve a problem.
Pseudocode	Plain English written in a code-like structure to plan an algorithm.

— End of Chapter 13 —

CHAPTER 14

ALGORITHMS

Computer Science — Beginner Level | Designed for First-Time Learners

Learning Objectives

- ✓ Define an algorithm and explain its importance in computing
- ✓ Identify the properties of a good algorithm
- ✓ Write simple algorithms using natural language and pseudocode
- ✓ Understand searching and sorting algorithms
- ✓ Trace through an algorithm step by step

14.1 What is an Algorithm?

An algorithm is a precise, step-by-step set of instructions for solving a problem or completing a task. Algorithms are the foundation of all computer programs — before writing a single line of code, a programmer must design an algorithm.

Did You Know?

The word "algorithm" comes from the name of the 9th-century Persian mathematician Al-Khwarizmi, whose works on arithmetic introduced systematic problem-solving methods to the world.

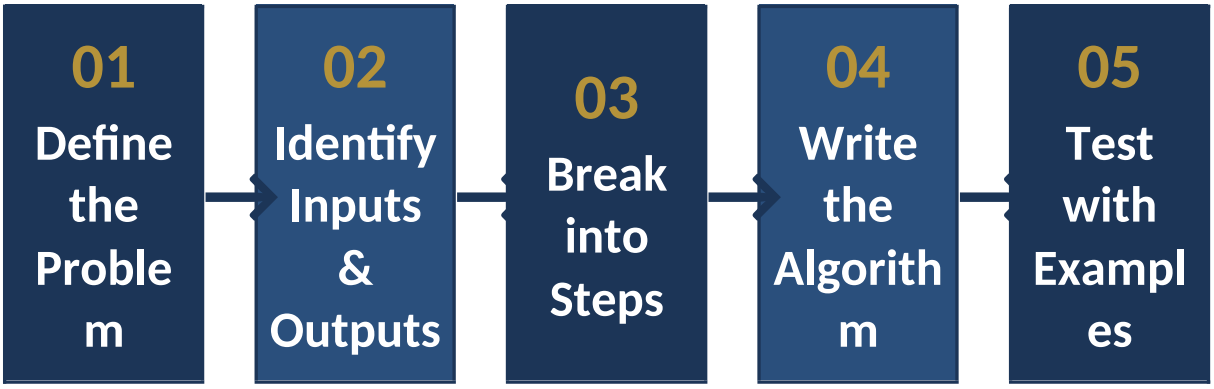


Figure 14.1: The five-step process for designing a good algorithm

14.2 Properties of a Good Algorithm

Property	Description
Correctness	Produces the right output for all valid inputs.
Clarity	Each step is clear and unambiguous.
Finiteness	The algorithm terminates after a finite number of steps.
Efficiency	Uses as little time and memory as possible.
Generality	Works for all valid inputs, not just specific cases.

14.3 Writing Algorithms: Pseudocode

Pseudocode is an informal way of writing algorithms using plain English structured like a programming language. It is a planning tool, not actual code.

Example: Find the Largest of Two Numbers

```
START
    INPUT number1
    INPUT number2
    IF number1 > number2 THEN
        OUTPUT "number1 is larger"
    ELSE
        OUTPUT "number2 is larger or they are equal"
    END IF
STOP
```

Example: Calculate Average of Three Numbers

```
START
    INPUT a
    INPUT b
    INPUT c
    SET total = a + b + c
    SET average = total / 3
    OUTPUT average
STOP
```

14.4 Searching Algorithms

Algorithm	Method	Best For
Linear Search	Check each item one by one from the start until found or list ends.	Small, unsorted lists
Binary Search	Start in the middle. If target is smaller go left, if larger go right. Repeat.	Large, sorted lists — much faster

Binary Search — Trace Example

Sorted list: [3, 7, 9, 11, 14, 25] Target: 25

- Step 1: Middle = index 2 (value 9). 25 > 9 → search right half [11, 14, 25].
- Step 2: Middle = index 4 (value 14). 25 > 14 → search right [25].
- Step 3: Only item = 25. FOUND.

14.5 Sorting Algorithms

Algorithm	Method	Efficiency
Bubble Sort	Repeatedly compare adjacent items and swap if out of order.	Simple but slow for large lists
Selection Sort	Find the smallest item and place it at the front. Repeat.	Simple, fewer swaps
Insertion Sort	Build a sorted portion by inserting each item in the correct position.	Good for small or nearly sorted lists

Bubble Sort — Trace Example

List: [5, 3, 8, 1]

Pass 1: [3,5,8,1] → [3,5,1,8] Pass 2: [3,1,5,8] Pass 3: [1,3,5,8] ✓

Chapter Review Questions

A. Short Answer Questions

- What is an algorithm? Why are algorithms important in computing?
- List the five properties of a good algorithm.
- What is pseudocode? How is it different from actual code?
- Explain the difference between linear search and binary search.
- Describe how bubble sort works. Trace it with [4, 2, 7, 1].

B. Fill in the Blanks

- An algorithm is a step-by-step set of _____ for solving a problem.
- _____ search requires the list to be sorted first.
- Bubble sort works by repeatedly comparing _____ items.
- An algorithm must _____ — it cannot run forever.

C. True or False

- Binary search works on unsorted lists. (True / False)
- Pseudocode must follow exact programming syntax. (True / False)
- An algorithm must produce the correct result for all valid inputs. (True / False)
- Linear search is more efficient than binary search for large sorted lists. (True / False)

Key Terms Glossary

Term	Definition
Algorithm	A precise, ordered set of instructions for solving a problem.
Pseudocode	Informal structured English used to plan an algorithm before coding.
Linear Search	Checking each item one by one from the start until found.
Binary Search	Efficient search on a sorted list by repeatedly halving the range.
Bubble Sort	Sorting by repeatedly comparing and swapping adjacent items.
Selection Sort	Sorting by repeatedly finding the minimum and placing it at the front.
Insertion Sort	Building a sorted portion by inserting each element correctly.
Trace	Manually stepping through an algorithm with example data.

— End of Chapter 14 —

CHAPTER 15

FLOWCHARTS

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Define a flowchart and explain its purpose
- ✓ Identify and correctly use standard flowchart symbols
- ✓ Draw flowcharts for simple and conditional processes
- ✓ Convert a written algorithm into a flowchart
- ✓ Trace through a flowchart to determine its output

15.1 What is a Flowchart?

A flowchart is a visual diagram that represents an algorithm or process using standard symbols connected by arrows. Flowcharts make it easier to understand, communicate, and check a process before writing code.

15.2 Standard Flowchart Symbols

Symbol	Shape	Used For
Terminal	Oval / Rounded Rectangle	Start and End of the process
Process	Rectangle	An action or instruction (e.g., "Add x to total")
Decision	Diamond	A yes/no question or condition (e.g., "Is x > 0?")
Input / Output	Parallelogram	Reading input or displaying output
Flow Arrow	Arrow / Line	Direction of flow between steps
Connector	Small Circle	Links parts of the flowchart across pages

■ Key Rule for Decision Symbols

Every decision diamond must have exactly two exits — one labelled YES (or TRUE) and one labelled NO (or FALSE). Always label both paths clearly.

15.3 Rules for Drawing Flowcharts

- Every flowchart must have exactly one START and at least one END.
- All paths must eventually lead to an END — no dead ends.
- Arrows show direction of flow — avoid unnecessary crossings.
- Use standard symbols consistently throughout.
- Decision symbols must have both exits labelled YES/NO.

15.4 Conditional Flowchart Example

Example: Check if a student has passed an exam (pass mark = 50)

Flowchart: Pass / Fail Check

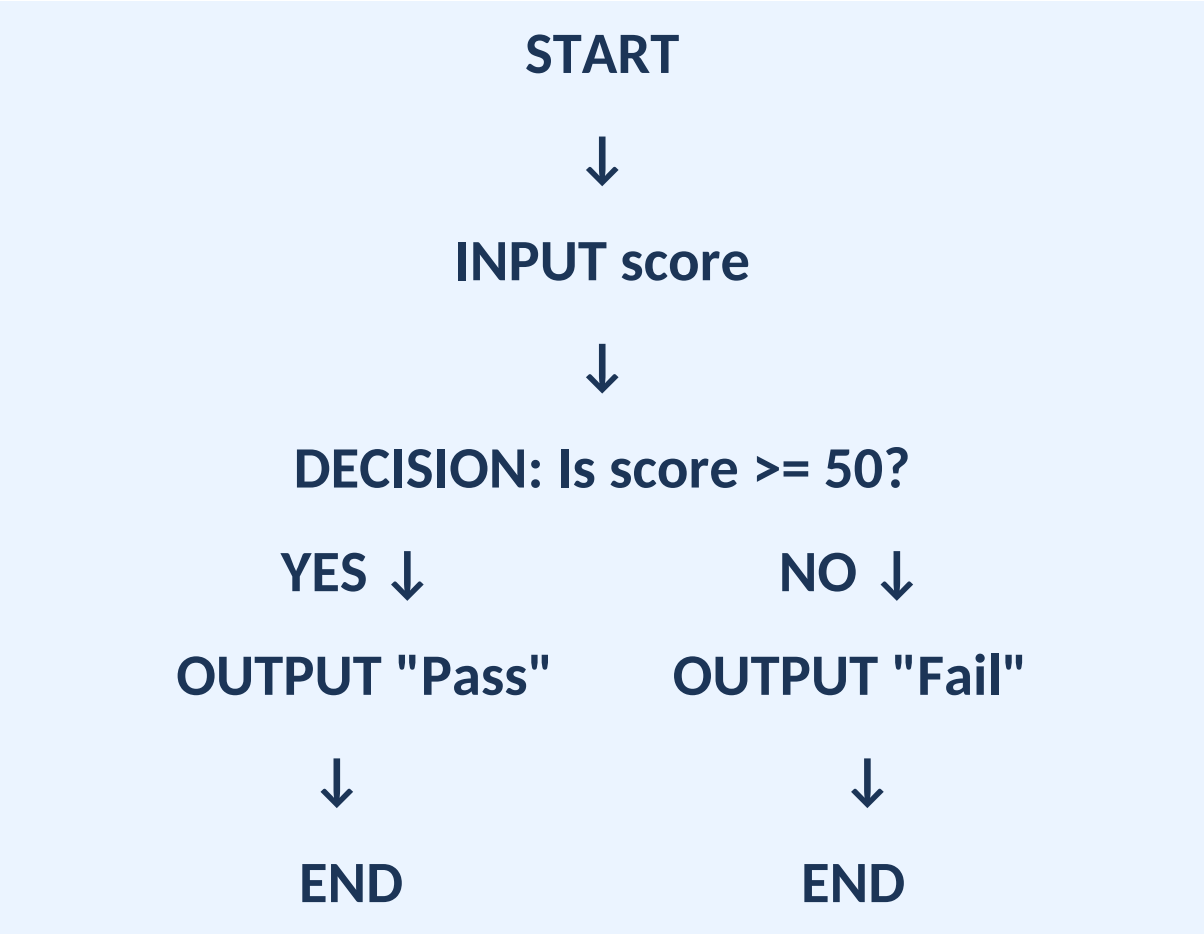


Figure 15.1: A conditional flowchart with a decision diamond

15.5 Loop in a Flowchart

A loop is created when an arrow points backward to an earlier part of the diagram. A decision diamond controls when the loop exits.

Example: Print numbers 1 to 5

START → SET n = 1 → OUTPUT n → Is n = 5? [YES] → END [NO] → SET n = n + 1 → (back to OUTPUT n)

15.6 Flowchart vs. Pseudocode

Feature	Flowchart	Pseudocode
Format	Visual / graphical diagram	Text-based structured English
Branching visibility	Immediately visible as separate paths	Written as IF/ELSE blocks
Loop visibility	Arrow bends back — very clear	WHILE/FOR loop written out
Best for	Non-programmers; simple logic	Complex logic; closer to actual code

Chapter Review Questions

A. Short Answer Questions

- What is a flowchart? When would you use one?
- Name and describe all five standard flowchart symbols.
- What must every decision symbol contain?
- What is the difference between a flowchart and pseudocode?

- Draw a flowchart that asks for a number and prints whether it is odd or even.

B. Fill in the Blanks

- A decision symbol is shaped like a _____ in a flowchart.
- Every decision symbol must have exactly _____ exits.
- The _____ symbol (oval) marks the start and end of a flowchart.
- A loop is created when an arrow points _____ to an earlier step.

C. True or False

- A flowchart can have multiple START symbols. (True / False)
- Decision symbols in a flowchart are shaped like rectangles. (True / False)
- All paths in a flowchart must eventually reach an END symbol. (True / False)
- A flow arrow shows the direction of logic flow. (True / False)

Key Terms Glossary

Term	Definition
Flowchart	A visual diagram representing an algorithm using standard symbols.
Terminal	Oval symbol marking the start or end of a flowchart.
Process	Rectangle symbol representing an action or instruction.
Decision	Diamond symbol representing a yes/no question or condition.
Input/Output	Parallelogram symbol for reading input or displaying output.
Flow Arrow	A line with an arrow showing the direction of logic flow.
Loop	A repeated section created when an arrow points back to an earlier step.

— End of Chapter 15 —

CHAPTER 16

INTRODUCTION TO CODING (SCRATCH)

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Explain what Scratch is and why it is used to learn programming
- ✓ Navigate the Scratch interface: Stage, Sprites, and Blocks palette
- ✓ Understand and use the main categories of Scratch blocks
- ✓ Create programs using events, loops, and conditional statements
- ✓ Use variables to store and change values in a Scratch project
- ✓ Build a simple interactive project in Scratch

16.1 What is Scratch?

Scratch is a free, visual programming language developed by MIT (Massachusetts Institute of Technology). Instead of typing lines of code, you build programs by snapping together colourful instruction blocks — making programming accessible to complete beginners. Access it free at: scratch.mit.edu

Scratch teaches sequences, loops, conditions, events, and variables — the same core ideas used in Python and all professional languages.

16.2 The Scratch Interface

Area	Location	Purpose
Stage	Top-left panel	The screen where your project runs.
Sprite List	Bottom-left panel	Shows all sprites. Click one to program it.
Blocks Palette	Middle panel	All available code blocks, organised by category.
Scripts Area	Right panel	Drag and connect blocks to build your code.
Green Flag	Top of Stage	Click to START your project.
Red Stop Button	Top of Stage	Click to STOP your project.

16.3 Block Categories

Category	Colour	What It Does
Motion	Dark Blue	Moves, rotates, and positions the sprite on the Stage.
Looks	Purple	Changes size, costumes, visibility; shows speech bubbles.
Sound	Pink	Plays sounds and music.
Events	Yellow	Triggers scripts (e.g., when green flag clicked).
Control	Orange	Loops (repeat, forever), conditions (if/else), wait.
Sensing	Light Blue	Detects touching, mouse position, keyboard input.
Operators	Green	Mathematical and logical operations (+, -, *, /, =, >, <).
Variables	Orange-Red	Creates and manages variables and lists.

16.4 Events

Event Block	When It Triggers
When green flag clicked	When the user presses the green flag to start.
When [key] pressed	When the user presses a specified keyboard key.
When this sprite clicked	When the user clicks directly on the sprite.
When I receive [message]	When a broadcast message is received from another sprite.

16.5 Loops in Scratch

Loop Block	Type	Use
repeat 10	Count-controlled	Runs the blocks inside exactly 10 times.
forever	Infinite	Runs the blocks inside continuously until stopped.
repeat until <condition>	Condition-controlled	Repeats until a condition becomes true.

16.6 Conditions and Variables

Conditions allow your sprite to make decisions:

if <score > 10> then → say "You win!" else → say "Keep trying!"

A variable is a named container that stores a value which can change — useful for tracking scores, lives, or timers.

- Click Variables > Make a Variable and give it a name (e.g., "score").
- New blocks appear: set score to [0], change score by [1].

16.7 Sample Mini-Project — Catch the Ball

- Basket sprite: controlled by left/right arrow keys using Motion blocks.
- Ball sprite: falls from the top using a forever loop with change y by -5.
- Variable "score" starts at 0, increases by 1 when ball touches basket.
- End condition: when score reaches 10, broadcast "game over" and stop all.

Chapter Review Questions

A. Short Answer Questions

- What is Scratch? Who developed it and where can you access it?
- Name four areas of the Scratch interface and what each does.
- What is a sprite? What is a backdrop?
- What are events in Scratch? Name three event blocks.
- What is the difference between a repeat loop and a forever loop?
- What is a variable in Scratch? Give an example of how to use one.

B. True or False

- You must type code to program in Scratch. (True / False)
- A sprite can have multiple costumes for animation. (True / False)
- The forever block runs its contents exactly 10 times. (True / False)
- Variables in Scratch can store values that change while the program runs. (True / False)

Key Terms Glossary

Term	Definition
Scratch	A free visual block-based programming language by MIT for beginners.
Sprite	A character or object in Scratch that can move and respond to events.
Stage	The display area where a Scratch project runs.
Backdrop	The background image displayed on the Stage.
Block	A single coloured instruction that connects to others to form a script.
Event	A trigger block that starts a script (e.g., when green flag clicked).
Loop	A block that repeats its contents (repeat, forever, repeat until).
Variable	A named container storing a changing value (e.g., score, lives).
Broadcast	A message sent from one sprite to trigger scripts in other sprites.

— End of Chapter 16 —

CHAPTER 17

BASIC PYTHON PROGRAMMING

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Understand what Python is and why it is widely used
- ✓ Write and run basic Python programs
- ✓ Use variables, data types, and operators
- ✓ Write conditional statements (if / elif / else)
- ✓ Use for and while loops
- ✓ Define and call functions
- ✓ Work with lists as basic data structures

17.1 What is Python?

Python is a high-level, general-purpose programming language known for its clean, readable syntax. Created by Guido van Rossum and first released in 1991, Python is now one of the most popular programming languages in the world — used in web development, data science, AI, automation, and education.

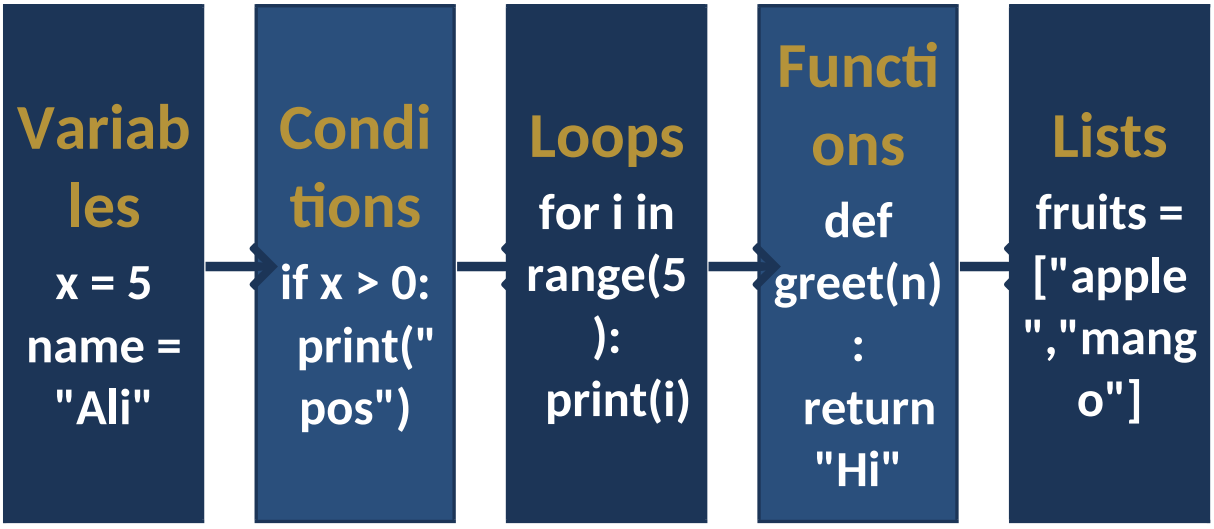


Figure 17.1: Core Python concepts — variables, conditions, loops, functions, and lists

17.2 Variables and Data Types

Data Type	Description	Example
int	Whole numbers (no decimal)	age = 17
float	Decimal numbers	height = 1.75
str	Text / characters (in quotes)	name = "Ali"
bool	True or False	is_student = True
list	An ordered collection of values	scores = [90, 85, 78]

17.3 Input and Output

```
print("Hello, World!")                    # Displays text on screen
name = input("What is your name? ")        # Gets text input from user
age  = int(input("How old are you? "))     # Converts input to integer
# Note: input() always returns a string — convert it for numbers!
```

17.4 Operators

Type	Operators	Examples
Arithmetic	+ - * / // % **	10 + 3 = 13 | 10 // 3 = 3 | 2**3 = 8
Comparison	== != < > <= >=	5 == 5 → True | 3 != 4 → True
Logical	and or not	(5>3) and (2<4) → True | not True → False
Assignment	= += -= *=	x = 5 | x += 3 (x becomes 8)

17.5 Conditional Statements (if / elif / else)

```
age = int(input("Enter your age: "))

if age >= 18:
    print("You are an adult.")
elif age >= 13:
    print("You are a teenager.")
else:
    print("You are a child.")
```

■ Indentation in Python

Python uses indentation (spaces at the start of a line) to define code blocks. Every line inside an if, elif, else, loop, or function must be indented consistently (usually 4 spaces). Missing indentation causes an IndentationError.

17.6 Loops

Loop Type	When to Use	Syntax
for loop	When you know how many times to repeat, or iterate over a list.	for i in range(5):
while loop	When you repeat until a condition becomes False.	while x > 0:

for Loop Examples

```
for i in range(5):          # prints 0, 1, 2, 3, 4
    print(i)

fruits = ["apple", "banana", "mango"]
for fruit in fruits:        # prints each fruit
    print(fruit)
```

while Loop Example

```
count = 1
while count <= 5:
    print(count)
    count += 1       # Always update variable to avoid infinite loop!
```

17.7 Functions

A function is a reusable block of code that performs a specific task.

```
def greet(name):
    message = "Hello, " + name + "!"
    return message

result = greet("Amina")
print(result)          # Output: Hello, Amina!
```

Term	Meaning	Example
def	Keyword to define a function.	def greet(name):
Parameter	Variable in the definition that receives a value.	name in greet(name)
Argument	Actual value passed when calling the function.	"Amina" in greet("Amina")
return	Sends a value back to the caller.	return message

17.8 Lists

A list is an ordered, changeable collection of values stored in a single variable.

```
fruits = ["apple", "banana", "mango"]
print(fruits[0])           # Output: apple   (index starts at 0)
fruits.append("orange")    # Add item to end
fruits.remove("banana")    # Remove specific item
print(len(fruits))         # Output: 3
print(fruits[-1])          # Output: mango   (last item)
```

Operation	Code	Result
Access by index	fruits[0]	First item ("apple")
Access last item	fruits[-1]	Last item
Add to end	fruits.append("orange")	Adds orange
Remove item	fruits.remove("banana")	Removes banana
Length	len(fruits)	Number of items
Sort	fruits.sort()	Alphabetical order

17.9 String Methods

Strings in Python have many built-in methods that make text processing easy. Call them using dot notation: string.method()

```
name = "Hello, World!"
```

```
print(name.upper())          # HELLO, WORLD!
print(name.lower())          # hello, world!
print(name.replace("World", "Python"))  # Hello, Python!
print(name.split(", "))      # ["Hello", "World!"]
print(len(name))             # 13
print(name.strip())          # removes leading/trailing spaces
print("Ali" in name)         # False  (checks if substring exists)
```

Method	What It Does
.upper()	Converts all characters to uppercase.
.lower()	Converts all characters to lowercase.
.strip()	Removes leading and trailing whitespace.
.replace(old, new)	Replaces all occurrences of old with new.
.split(sep)	Splits the string at sep and returns a list.
.find(sub)	Returns the index of sub, or -1 if not found.
len(string)	Returns the number of characters (built-in function).
in	Checks if a substring exists: "Ali" in name → True/False

17.10 Dictionaries

A dictionary stores data as key-value pairs. Each key is unique and maps directly to its value — like a real dictionary where each word (key) maps to its definition (value).

```
student = {
    "name"  : "Amina",
    "age"   : 16,
    "grade" : "A"
}

print(student["name"])       # Amina
student["age"] = 17          # Update a value
student["city"] = "Cairo"    # Add a new key-value pair
del student["grade"]         # Delete a key

for key, value in student.items():
    print(key, ":", value)   # Loop through all pairs
```

Operation	Code	Result
Access value	student["name"]	"Amina"
Add / Update	student["age"] = 17	Updates age to 17
Delete key	del student["grade"]	Removes that pair
Check if key exists	"name" in student	True or False
Get all keys	student.keys()	All key names
Get all values	student.values()	All value data
Loop all pairs	student.items()	(key, value) tuples

17.11 Error Handling (try / except)

Errors are normal in programming. Python uses try/except blocks to handle errors gracefully — preventing the program from crashing and showing the user a helpful message instead.

```
try:
    number = int(input("Enter a number: "))
    result = 100 / number
    print("Result:", result)

except ValueError:
    print("Error: Please enter a valid number.")

except ZeroDivisionError:
    print("Error: You cannot divide by zero.")

finally:
    print("Program finished.")  # Always runs
```

Block	Purpose
try	Contains the code that might cause an error. Python attempts to run it.
except ErrorType	Runs if the specified error occurs. Handles it without crashing.
except (multiple)	You can have multiple except blocks for different error types.
finally	Runs whether an error occurred or not — good for cleanup actions.

Did You Know?

Common Python error types: ValueError (wrong data type given), ZeroDivisionError (dividing by zero), IndexError (list index out of range), FileNotFoundError (file does not exist), and TypeError (wrong type for an operation).

Chapter Review Questions

A. Short Answer Questions

- What is Python? Name three professional fields where it is used.
- What is the difference between int, float, str, and bool?
- What does input() return? How do you convert it to a number?
- What is the difference between a for loop and a while loop?
- What is a function? Why are functions useful?
- What is a list? How do you access the third item?
- Name four string methods and explain what each does.
- What is a dictionary? How is it different from a list?
- What is error handling? Why is try/except important?

B. What is the Output?

- print(2 ** 3)
- print(10 % 3)
- x = 5; x += 3; print(x)
- for i in range(2, 8, 2): print(i)
- fruits = ["apple", "banana", "cherry"]; print(fruits[-1])

C. Write the Python Code

- Write a program that asks for two numbers and prints their sum, difference, and product.
- Write a function called is_even(n) that returns True if n is even and False if odd.
- Write a program that prints the multiplication table of a number entered by the user.

Key Terms Glossary

Term	Definition
Python	A high-level, readable programming language used in many fields.
Variable	A named storage location holding a value.
Data Type	The kind of value a variable holds (int, str, float, bool, list, dict).
print()	Built-in function to display output on screen.
input()	Built-in function to receive text input from the user.
for loop	A loop that repeats a set number of times or iterates over a sequence.
while loop	A loop that repeats as long as a condition remains True.
Function	A reusable block of code defined with def.
List	An ordered, changeable collection of values stored in one variable.
Dictionary	A collection of key-value pairs — each key maps to a specific value.
String Method	A built-in function called on a string (e.g., .upper(), .split()).
try / except	A block that handles errors gracefully without crashing the program.
Indentation	Spaces at the start of a line that define code blocks in Python.
return	Sends a value back from a function to the caller.
ValueError	Error raised when a function receives an argument of the wrong type.

— End of Chapter 17 —

CHAPTER 18

ARTIFICIAL INTELLIGENCE

Computer Science — Beginner Level | Designed for First-Time Learners

Learning Objectives

- ✓ Define Artificial Intelligence and explain its key concepts
- ✓ Describe different types of AI and machine learning
- ✓ Identify real-world applications of AI in everyday life
- ✓ Understand how AI learns from data (supervised, unsupervised, reinforcement)
- ✓ Discuss the benefits and limitations of current AI systems

18.1 What is Artificial Intelligence?

Artificial Intelligence (AI) is the branch of computer science concerned with creating systems that can perform tasks that would normally require human intelligence. These tasks include understanding language, recognising images, making decisions, solving problems, and learning from experience.

The term "Artificial Intelligence" was coined by computer scientist John McCarthy in 1956. Today, AI is one of the most rapidly advancing fields in technology.

18.2 Types of AI

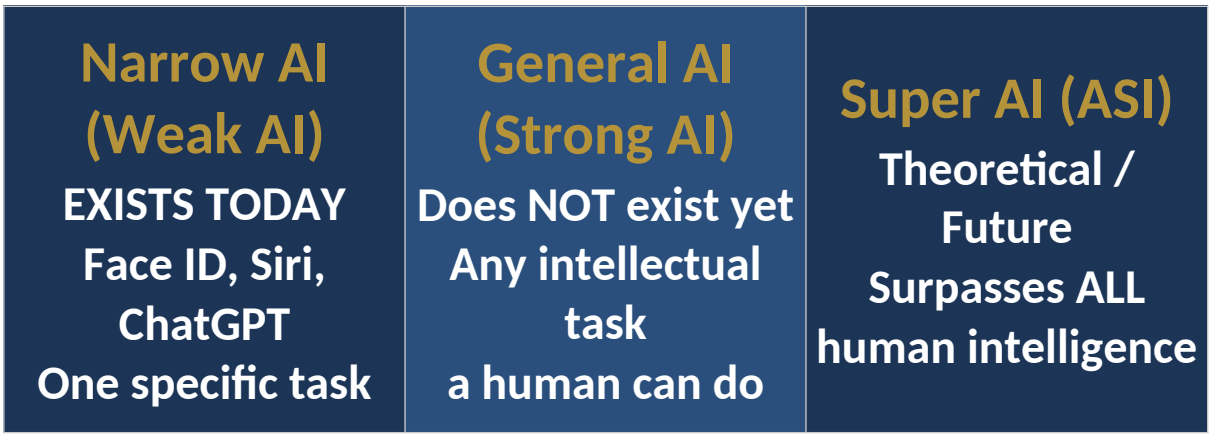


Figure 18.1: The three types of AI — only Narrow AI currently exists

Type	Description	Status
Narrow AI (Weak AI)	AI designed for one specific task. Cannot do anything outside its training. Examples: face recognition, spam filters, chatbots.	EXISTS TODAY
General AI (Strong AI)	Hypothetical AI that can perform any intellectual task a human can. Would reason across any domain.	Does NOT exist yet
Super AI (ASI)	Hypothetical AI that surpasses human intelligence in every domain.	Theoretical — far future

18.3 Machine Learning

Machine Learning (ML) is the most important technique in modern AI. Instead of writing explicit rules, we feed an ML system data and it learns the rules by itself by finding patterns.

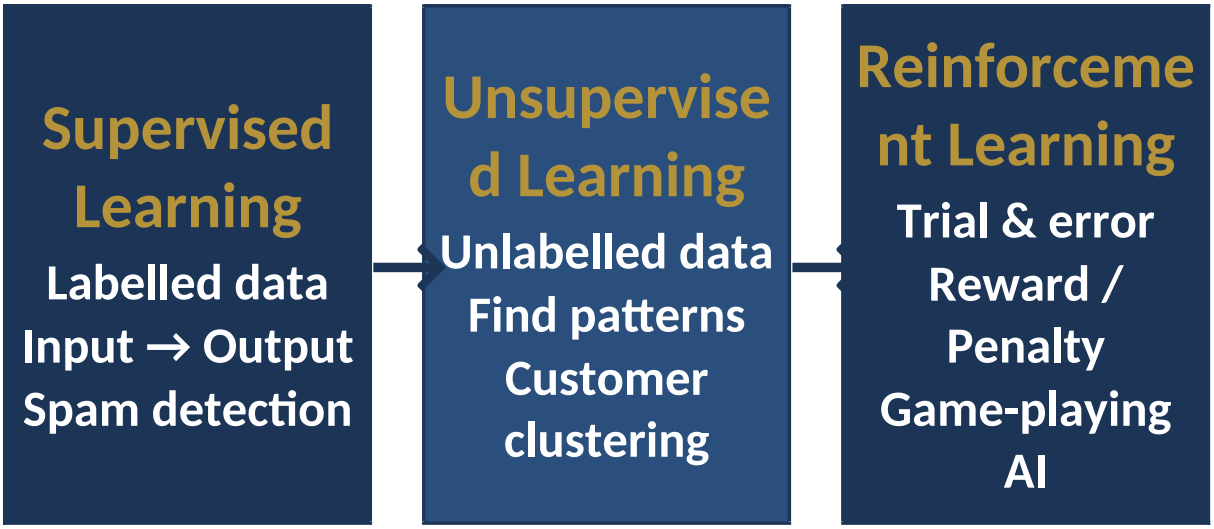


Figure 18.2: The three main types of machine learning

Type	Description	Examples
Supervised Learning	Training data is labelled — inputs and correct outputs are both given.	Spam detection, image classification
Unsupervised Learning	Training data is NOT labelled. The model finds its own patterns.	Customer segmentation, anomaly detection
Reinforcement Learning	The AI learns by trial and error — rewards for correct actions, penalties for wrong ones.	Game-playing AI (chess, Go), robot navigation

Neural Networks and Deep Learning

A neural network is a computing model inspired by the human brain. It is made up of layers of nodes (neurons) connected by weighted links. Data passes through the input layer, through hidden layers, and out through the output layer.

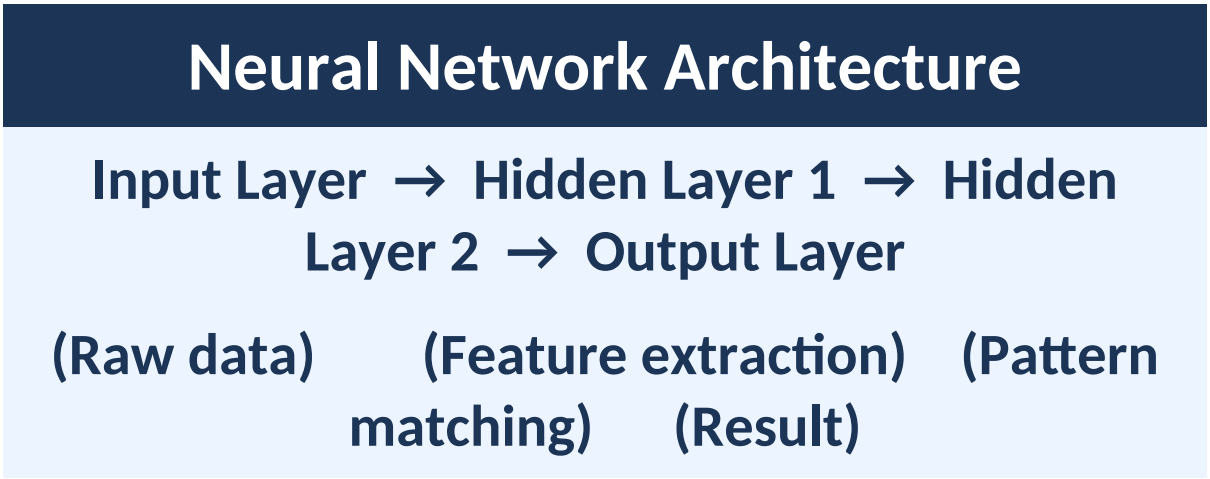


Figure 18.3: A neural network — deep learning uses many hidden layers

18.4 Natural Language Processing (NLP)

Natural Language Processing (NLP) is the field of AI that enables computers to understand, interpret, and generate human language — both written and spoken. It bridges the gap between human communication and machine understanding.

NLP Application	How It Works	Example
Voice Assistants	Converts speech to text, interprets intent, responds.	Siri, Alexa, Google Assistant
Machine Translation	Analyses sentence structure in one language, generates output in another.	Google Translate
Chatbots	Understands typed questions and generates relevant responses.	ChatGPT, customer service bots
Sentiment Analysis	Detects whether text is positive, negative, or neutral.	Analysing product reviews
Autocomplete / Autocorrect	Predicts next word or corrects spelling based on context.	Smartphone keyboard
Email Spam Filter	Reads email content and classifies as spam or not spam.	Gmail spam detection
Text Summarisation	Condenses long documents into short summaries automatically.	News summary apps

18.5 Computer Vision

Computer Vision is the field of AI that enables computers to interpret and understand images and video — in the same way a human uses their eyes and brain. It combines cameras, deep learning, and neural networks.

Computer Vision Application	Description	Real-World Use
Image Classification	Identifies what category an image belongs to.	Is this photo a cat or a dog?
Object Detection	Finds and labels multiple objects within an image.	Self-driving cars spotting pedestrians
Facial Recognition	Identifies a specific individual from their face.	Face ID on smartphones, security cameras
Medical Imaging Analysis	Detects abnormalities in X-rays, MRI, and CT scans.	Early cancer detection
Optical Character Recognition (OCR)	Converts text in images to editable digital text.	Scanning printed documents
Gesture Recognition	Reads hand or body movements as input commands.	Gaming controllers, sign language AI

18.6 AI Applications in Real Life

Field	AI Application
Healthcare	Medical image analysis (detecting cancer in X-rays), drug discovery, diagnosis support.
Transportation	Self-driving cars (Tesla, Waymo), GPS route optimisation, traffic management.
Education	Personalised learning platforms, automated essay grading, AI tutors.
Entertainment	Recommendation engines (Netflix, Spotify, YouTube), game AI.
Finance	Fraud detection, algorithmic trading, credit scoring, chatbot customer service.
Language	Machine translation (Google Translate), voice assistants (Siri, Alexa).
Security	Cybersecurity threat detection, biometric authentication (face ID).

Field	AI Application
Agriculture	Crop disease detection from drone images, automated harvesting robots.

18.7 Limitations of AI

Limitation	Explanation
Narrow Scope	Current AI only works in the specific area it was trained on.
Needs Large Data	ML models need huge amounts of high-quality training data.
Bias	If training data is biased, the AI will be biased too.
Explainability	Deep learning models are "black boxes" — hard to explain their decisions.
No Common Sense	AI lacks real-world understanding and general reasoning ability.
Energy Consumption	Training large AI models requires vast amounts of electricity.
Fragility	AI can fail when it encounters data outside its training distribution.

Chapter Review Questions

A. Short Answer Questions

- What is Artificial Intelligence? Define it in your own words.
- What is the difference between Narrow AI, General AI, and Super AI?
- Explain supervised, unsupervised, and reinforcement learning with one example each.
- What is a neural network? What is deep learning? How are they related?
- What is NLP? Name four real-world applications of NLP.
- What is Computer Vision? Give three examples of how it is used today.
- Name five real-world applications of AI across different fields.

B. Fill in the Blanks

- All current AI systems are examples of _____ AI.
- _____ learning trains models using labelled examples.
- NLP stands for Natural Language _____.
- Computer Vision allows computers to understand _____ and video.
- _____ analysis detects whether text is positive, negative, or neutral.
- AI training data that is _____ can lead to unfair or discriminatory results.

C. True or False

- General AI exists and is used in most smartphones today. (True / False)
- Reinforcement learning uses a reward system to train AI. (True / False)
- Deep learning models are always easy to explain. (True / False)
- Machine Learning systems learn from data rather than explicit rules. (True / False)
- Computer Vision is used in medical imaging to detect diseases. (True / False)
- Chatbots are an example of Natural Language Processing. (True / False)

Key Terms Glossary

Term	Definition
Artificial Intelligence	Computer systems that simulate human-like intelligence.

Machine Learning	AI that learns from data without being explicitly programmed.
Deep Learning	ML using multi-layered neural networks for complex tasks.
Neural Network	A computing model inspired by the human brain's layered structure.
NLP	Natural Language Processing — AI for understanding and generating human language.
Computer Vision	AI for interpreting and analysing images and video.
Supervised Learning	ML trained on labelled data (input + correct output provided).
Unsupervised Learning	ML trained on unlabelled data to find patterns independently.
Reinforcement Learning	ML trained through reward and penalty feedback.
Narrow AI	AI designed for one specific task — all current AI is Narrow AI.
Sentiment Analysis	NLP technique that determines whether text is positive, negative, or neutral.
Image Classification	Computer Vision task of identifying which category an image belongs to.
OCR	Optical Character Recognition — AI that converts text in images to digital text.
Facial Recognition	Computer Vision that identifies individuals from images of their face.

— End of Chapter 18 —

CHAPTER 19

ETHICS OF AI

Computer Science — Beginner Level | Designed for First-Time Learners

■ Learning Objectives

- ✓ Understand what AI ethics means and why it matters
- ✓ Identify key ethical issues: bias, privacy, fairness, transparency
- ✓ Discuss the social impact of AI on employment and society
- ✓ Understand the concept of responsible AI
- ✓ Evaluate AI decisions from an ethical perspective
- ✓ Articulate personal views on the future of AI

19.1 Why Ethics Matters in AI

Artificial Intelligence is becoming part of nearly every aspect of modern life — from the videos recommended to you, to decisions about who gets a bank loan, to medical diagnosis. As AI becomes more powerful and widespread, the decisions it makes have real consequences for real people.

AI ethics is the field that asks: How should AI be designed, deployed, and governed to ensure it is safe, fair, and beneficial for everyone?

Why Ethics Cannot Be Ignored

- AI systems have already wrongly identified innocent people as criminals (facial recognition bias).
- AI tools have denied loans or jobs unfairly due to biased training data.
- Deepfake AI technology has been used to spread misinformation at scale.
- Autonomous weapons are being developed that can make life-or-death decisions.

19.2 Key Principles of Ethical AI

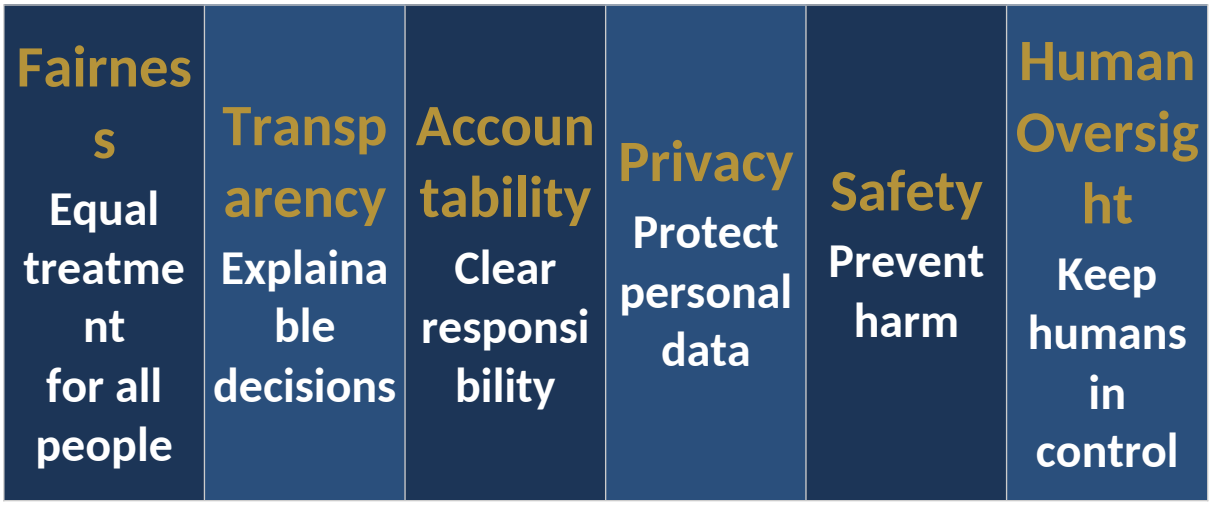


Figure 19.1: The six key principles of ethical AI development

Principle	Description
Fairness	AI should treat all people equally, regardless of race, gender, age, or religion.

Principle	Description
Transparency	AI decisions should be explainable. People should understand why an AI made a decision.
Accountability	When AI causes harm, there must be clear responsibility — developer, company, or user.
Privacy	AI should collect and use only the data necessary for its purpose.
Safety	AI must be thoroughly tested to ensure it does not cause harm.
Human Oversight	Humans must remain in control of important AI decisions.

19.3 AI Bias

AI bias occurs when an AI system produces unfair or discriminatory results. Bias usually enters AI through training data — if the data reflects historical inequalities, the AI will learn and perpetuate those inequalities.

Real-World Example	Bias Found
Amazon Hiring Algorithm (2018)	Penalised CVs mentioning women's colleges because it was trained on historically male-dominated data.
Facial Recognition Systems	Studies show some systems are far more accurate for white male faces than darker-skinned or female faces.
COMPAS (Criminal Justice USA)	A recidivism prediction tool used in US courts was found to show racial bias.
Healthcare Diagnostics	AI trained on data from one ethnic group may be less accurate for others.

How to Reduce AI Bias

- Use diverse, representative training data.
- Audit AI systems regularly for discriminatory outputs.
- Include diverse teams in AI development.
- Test AI on edge cases and minority groups.
- Make the decision-making process transparent and explainable.

19.4 Privacy and Surveillance

- Data Collection: Apps collect location, messages, purchases, and browsing history — often without users fully understanding what is shared.
- Facial Recognition: AI-powered cameras can identify individuals in public without their knowledge or consent.
- Profiling: AI can build detailed profiles of individuals to predict or influence behaviour.
- Data Breaches: If personal data used to train AI is hacked, people's privacy is violated at scale.

■ GDPR — Data Protection Law

The EU's General Data Protection Regulation (GDPR) gives people rights over their personal data — including the right to know what data is collected, the right to have it deleted, and the right not to be subject to fully automated decisions that significantly affect them.

19.5 Deepfakes and Misinformation

Risk	Description
Political misinformation	Fake videos of politicians saying things they never said — used to influence elections.

Risk	Description
Non-consensual images	A serious form of online abuse causing severe harm to victims.
Fraud	Fake audio of a CEO's voice used to trick companies into transferring money.
Erosion of trust	When nothing can be trusted as genuine, democratic processes suffer.

19.6 AI and Employment

Jobs at Risk from AI	Jobs Created by AI
Routine data entry and processing	AI engineers and researchers
Basic customer service (call centres)	Data scientists and analysts
Some transport and delivery (self-driving)	AI ethics officers and policy advisors
Simple legal document review	AI trainers and data labellers
Basic financial and accounting tasks	Cybersecurity specialists

■ The Key Skill for the Future

Humans working alongside AI — using creativity, emotional intelligence, and critical thinking that machines cannot replicate.

The question is not "Will AI replace me?" but "How can I use AI as a tool?"

19.7 Responsible AI Development

Organisation	AI Ethics Initiative
European Union	EU AI Act (2024) — the first comprehensive AI regulation law in the world.
UNESCO	Recommendation on the Ethics of AI — global framework adopted by 193 countries.
Google	AI Principles — guidelines for responsible AI development and deployment.
Microsoft	Responsible AI Standard — fairness, reliability, privacy, inclusiveness, transparency, accountability.
Anthropic	Constitutional AI — a framework for training AI to be helpful, harmless, and honest.

19.8 Should We Be Worried About AI?

Perspective	Key Arguments
Optimistic View ●	AI will solve major global challenges (disease, climate, poverty). AI will free humans from dangerous and difficult tasks. AI will make education, healthcare, and services more accessible.
Cautious View ●	AI development should be slowed to allow safety research to catch up. More regulation and oversight is needed now. The benefits of AI must be shared fairly.
Critical View ●	AI could concentrate power in the hands of a few companies and governments. Advanced AI could pose existential risks if not developed carefully. Open public debate is essential.

For Discussion

- Do you think AI will make the world better or worse? Justify your view.
- What one rule would you put in place to make AI safer or fairer?
- Who should decide how AI is used — governments, companies, or the public?
- Is it ethical to use AI to make life-or-death medical or legal decisions?

Chapter Review Questions

A. Short Answer Questions

- What is AI ethics? Why does it matter for society?
- What is AI bias? Give two real-world examples and explain their harm.
- What is a deepfake? What are three risks it poses to society?
- How is AI changing employment? Name three jobs at risk and three new jobs.
- What are the key principles of responsible AI? Name and explain four.

B. Discussion Questions

- A hospital uses AI to decide which patients receive organ transplants. What ethical issues does this raise?
- Should facial recognition cameras be used in public spaces? Argue both for and against.
- Is it ethical for companies to use AI to screen job applicants? What safeguards should be required?

C. True or False

- AI bias only occurs if the programmers are personally biased. (True / False)
- Deepfakes are created using deep learning technology. (True / False)
- The EU AI Act is the first comprehensive AI regulation law in the world. (True / False)
- AI is expected to create no new jobs — only destroy existing ones. (True / False)
- Human oversight of AI decisions is especially important in high-stakes areas. (True / False)

Key Terms Glossary

Term	Definition
AI Ethics	The study of moral principles guiding how AI is designed, used, and governed.
AI Bias	Unfair or discriminatory outputs from an AI system, caused by biased training data.
Deepfake	AI-generated realistic fake image or video making it appear someone said something false.
Transparency	The ability to explain and understand how an AI system reaches its decisions.
Accountability	Being responsible for the consequences of an AI system's actions.
GDPR	EU regulation giving individuals rights over their personal data.
Autonomous Weapons	Weapons that can identify and attack targets without human control.
Responsible AI	AI developed and deployed with fairness, safety, privacy, and human welfare in mind.
GAN	Generative Adversarial Network — deep learning architecture used to create deepfakes.

EU AI Act	The world's first comprehensive AI regulation law, passed by the EU in 2024.

— End of Chapter 19 —

Foundations of Computer Science

From Basics to Artificial Intelligence

19 Comprehensive Chapters · 6 Units · Complete Edition

Unit	Chapters	Topics Covered
Unit 1: Computer Fundamentals	Ch. 1 – 6	Introduction, Hardware, Software, Data Storage, Operating Systems, File Management
Unit 2: Productivity & Office Tools	Ch. 7 – 9	MS Word, MS PowerPoint, MS Excel
Unit 3: Internet, Communication & Safety	Ch. 10 – 12	Internet Basics, Email & Communication, Cyber Safety & Etiquette
Unit 4: Computational Thinking	Ch. 13 – 15	Logical Thinking, Algorithms, Flowcharts
Unit 5: Introduction to Coding	Ch. 16 – 17	Scratch Programming, Basic Python Programming
Unit 6: Artificial Intelligence & Ethics	Ch. 18 – 19	Artificial Intelligence, Ethics of AI

— The Monotheist —

Author · Educator · Lifelong Student

" Use this knowledge well. Be honest. Be curious. Be kind to others in digital spaces as you would be in person. Build things that help people. Question things that harm them. And never stop learning. "